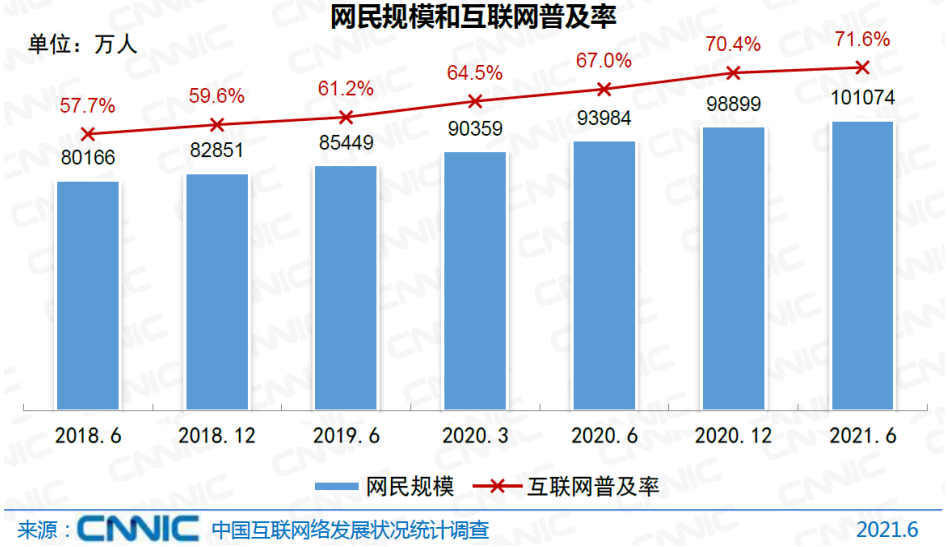
Computer Networks

Spring 2022

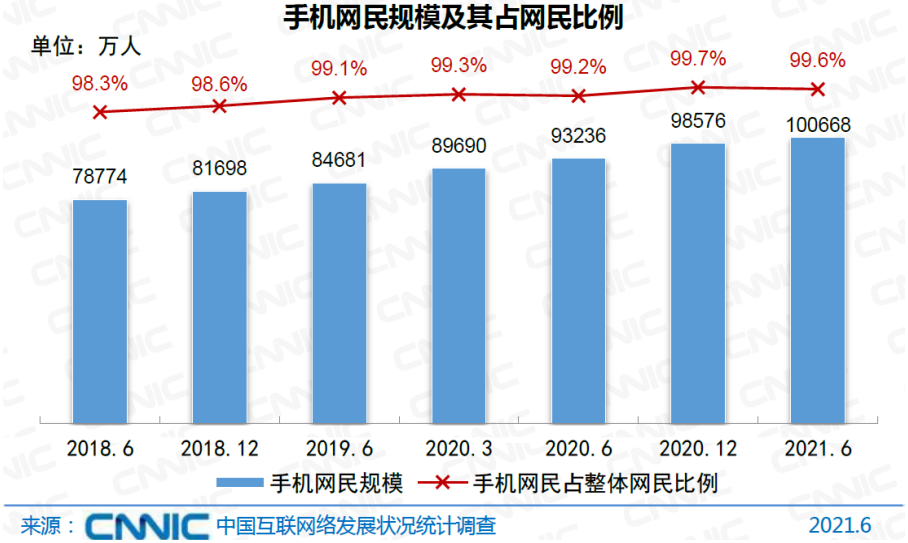


- Introduction: course description (Why to learn? What to learn? How to learn?)
- Answers to frequently asked questions


网民规模和互联网普及率
单位：万人
22.6%
28.9%
34.3%
38.3%
42.1%
45.8%
47.9%
50.3%
53.2%
55.8%
59.6%
29800
38400
45730
51310
56400
61758
64875
68826
73125
77198
82851
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
网民数
互联网普及率
来源：CNNIC中国互联网络发展状况统计调查
2018.12


网民规模和互联网普及率
单位：万人
57.7%
59.6%
61.2%
64.5%
67.0%
70.4%
71.6%
80166
82851
85449
90359
93984
98899
101074
2018. 6
2018. 12
2019. 6
2020. 3
2020. 6
2020. 12
2021. 6
网民规模
互联网普及率
来源：CNNIC 中国互联网络发展状况统计调查
2021.6


手机网民规模及其占网民比例
单位：万人
39.5%
60.8%
66.2%
69.3%
74.5%
81.0%
85.8%
90.1%
95.1%
97.5%
98.6%
11760
23344
30274
35558
41997
50006
55678
61981
69531
75265
81698
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
手机网民规模
手机网民占整体网民比例
来源：CNNIC中国互联网络发展状况统计调查
2018.12


手机网民规模及其占网民比例
单位：万人
98.3%
98.6%
99.1%
99.3%
99.2%
99.7%
99.6%
78774
81698
84681
89690
93236
98576
100668
2018. 6
2018. 12
2019. 6
2020. 3
2020. 6
2020. 12
2021. 6
手机网民规模
手机网民占整体网民比例
来源：CNNIC 中国互联网络发展状况统计调查
2021.6

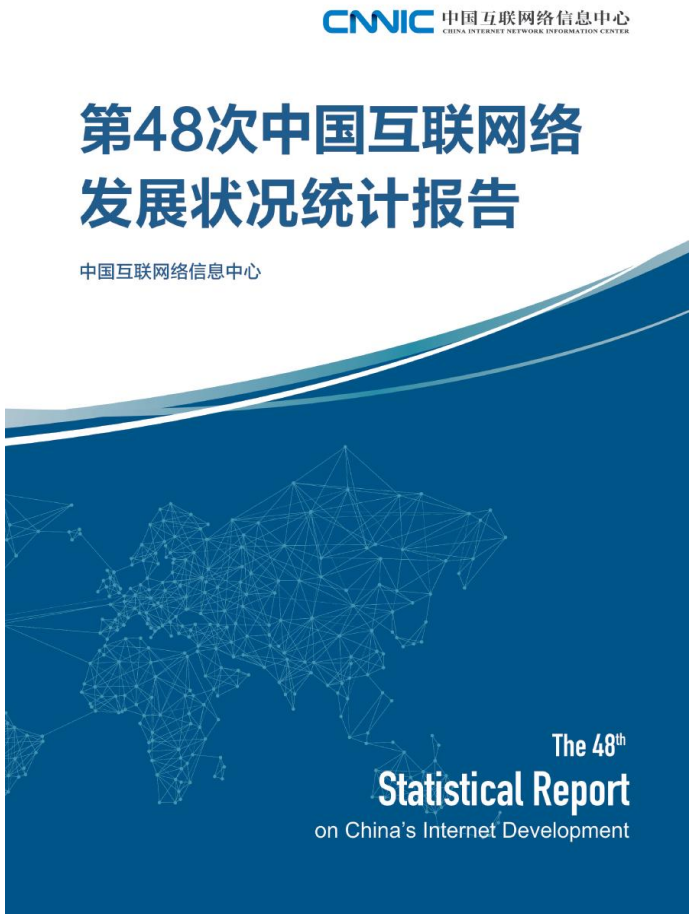
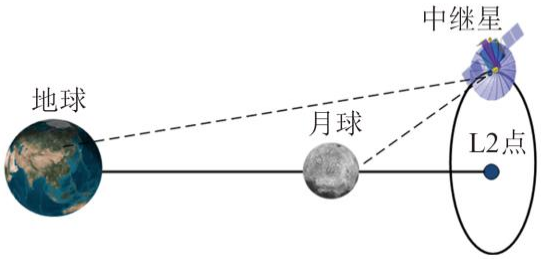


表 4　2020.12-2021.6 各类互联网应用用户规模和网民使用率

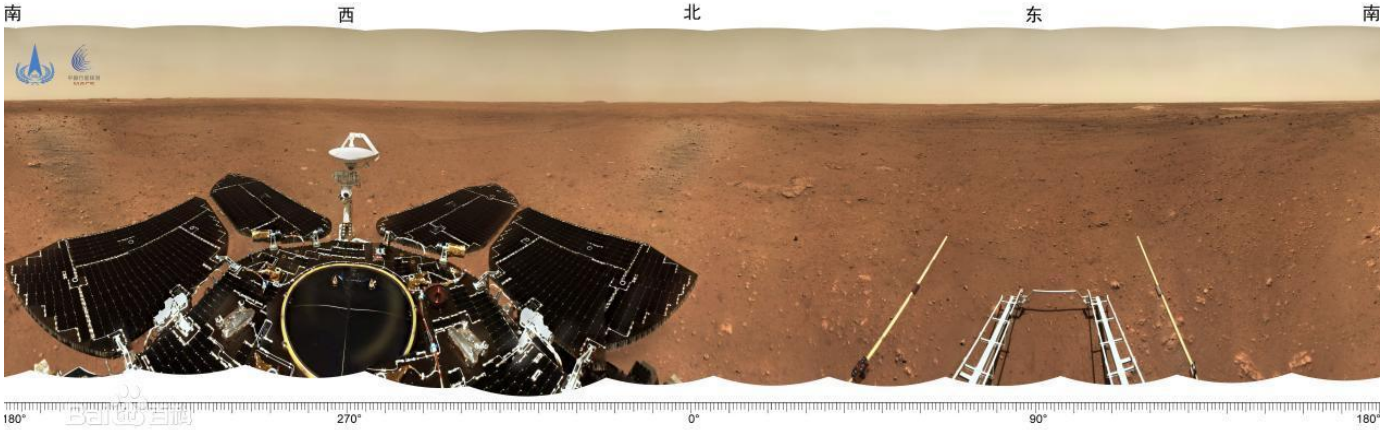
	2020.12		2021.6		
应用	用户规模（万）	网民使用率	用户规模（万）	网民使用率	增长率
即时通信	98111	99.2%	98330	97.3%	0.2%
网络视频（含短视频）	92677	93.7%	94384	93.4%	1.8%
短视频	87335	88.3%	88775	87.8%	1.6%
网络支付	85434	86.4%	87221	86.3%	2.1%
网络购物	78241	79.1%	81206	80.3%	3.8%
搜索引擎	76977	77.8%	79544	78.7%	3.3%
网络新闻	74274	75.1%	75987	75.2%	2.3%
网络音乐	65825	66.6%	68098	67.4%	3.5%
网络直播	61685	62.4%	63769	63.1%	3.4%
网络游戏	51793	52.4%	50925	50.4%	-1.7%
网上外卖	41883	42.3%	46859	46.4%	11.9%
网络文学	46013	46.5%	46127	45.6%	0.2%
网约车	36528	36.9%	39651	39.2%	8.5%
在线办公	34560	34.9%	38065	37.7%	10.1%
在线旅行预订	34244	34.6%	36655	36.3%	7.0%
在线教育	34171	34.6%	32493	32.1%	-4.9%
在线医疗	21480	21.7%	23933	23.7%	11.4%
互联网理财	16988	17.2%	16623	16.4%	-2.1%



通过鹊桥传回的月球背面图像



鹊桥中继星的作用

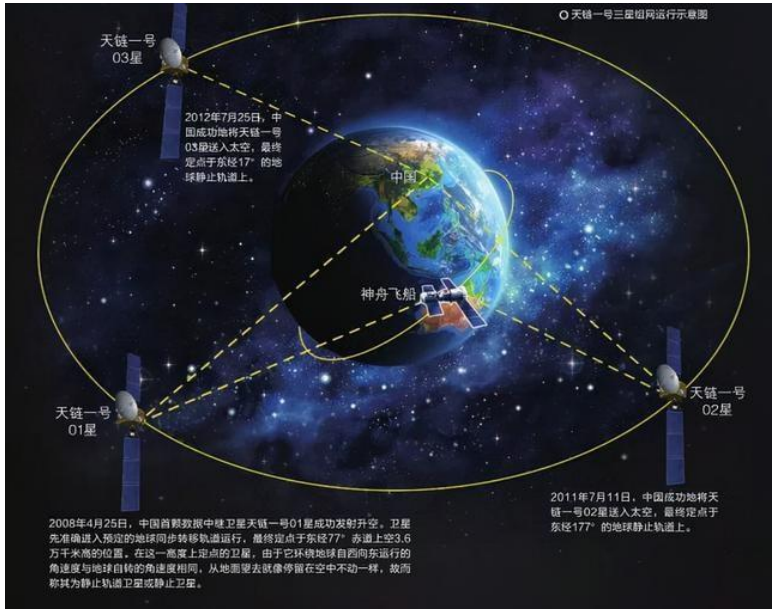


祝融号火星车传回的图像



“天宫课堂”第一课开讲





天链一号实现全球无缝链接通信



Metaverse一词在2021年12月，入选《柯林斯词典》2021年度热词；
12月6日，入选“2021年度十大网络用语”；
12月8日，入选《咬文嚼字》“2021年度十大流行语”。



- Introduction: course description (Why to learn? What to learn? How to learn?)
- Answers to frequently asked questions

Textbook

1. Computer Networking: A Top Down Approach, 7th ed., Pearson/Addison Wesley
2. 计算机网络：自顶向下方法，原书第7版，机械工业出版社
3. 何顶新，郑定富，胡晓娅. *计算机网络及应用*，北京：化学工业出版社，2011.06

Course Grading

- Assignments (experiment + homework ）+ attendance 30%
- Final 70%

Chapter 1: introduction

Overview:

- Terminology and Concepts related to
  " Computer Network"
  - ✓ use Internet as example
- What's a protocol?
- Protocol layers and their service models
- Internet history

Keypoints:

- Protocol and three essential factors
- Protocol layers and layered architecture
- 协议层和分层结构

Difficulties:

- Connection-oriented and connectionless
- Circuit switching and packet switching
- 电路交换和分组交换
- Delay and how to calculate the delay
- 延迟和如何计算延迟

Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

1.5 Protocol layers, Service models

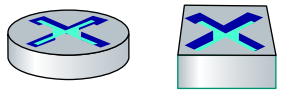
1.6* Internet history

What's the Internet: "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers, switches*

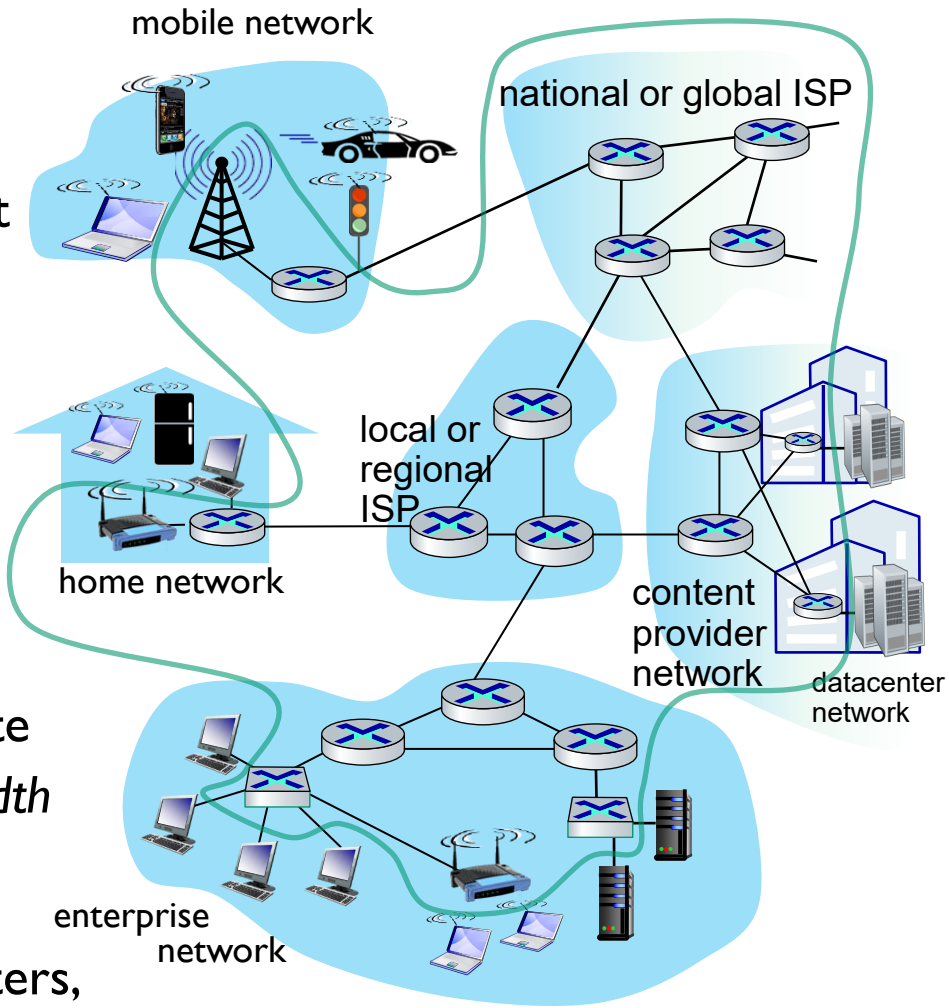


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



"Fun" Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



scooters



cars

bikes





Security Camera



Slingbox: remote control cable TV



Internet phones



Fitbit



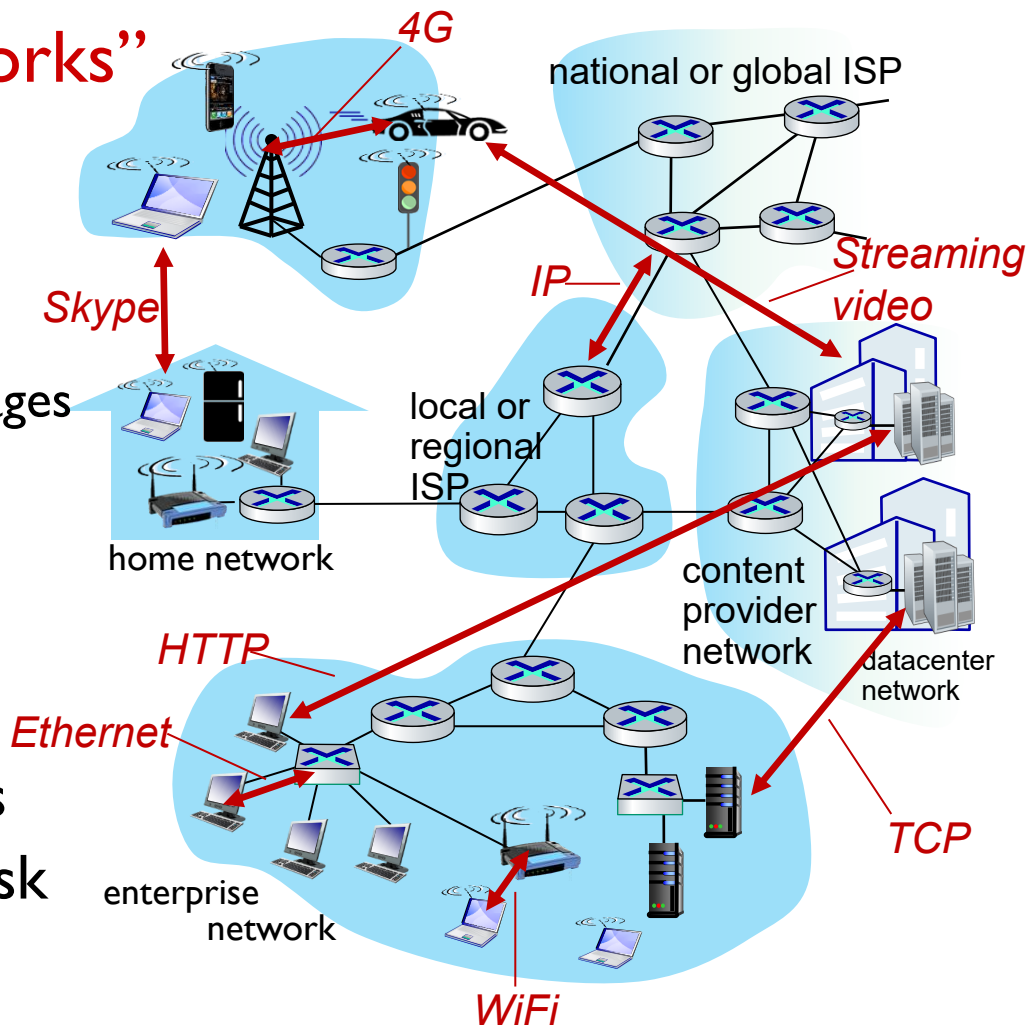
AR devices



Tweet-a-watt: monitor energy use

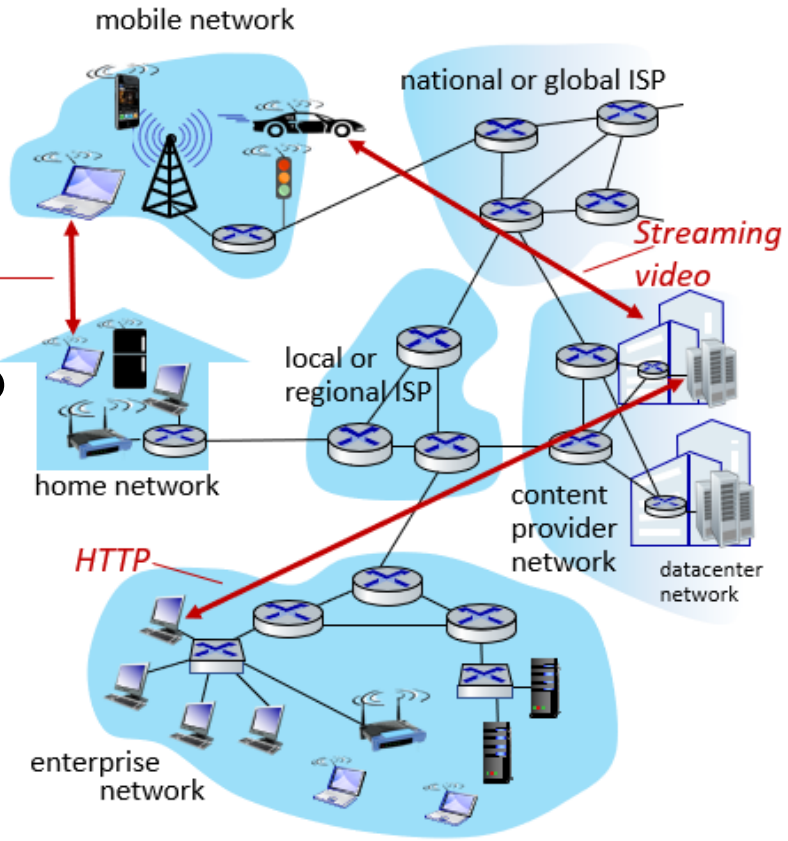
What's the Internet: "nuts and bolts" view

- *Internet*: "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), TCP, IP, WiFi, streaming video, Skype, 4G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



What's the Internet: a service view

- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …
- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, **analogous to postal service**



Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

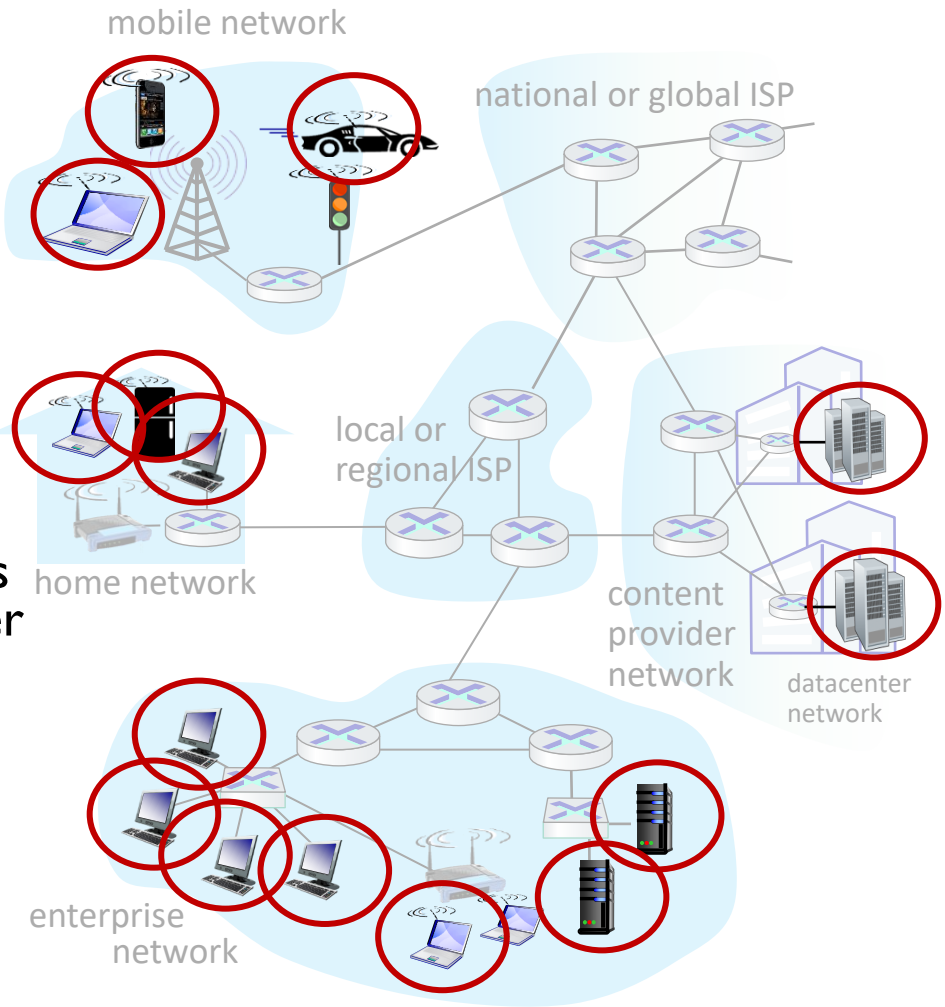
1.5 Protocol layers, Service models

1.6* Internet history

A closer look at network structure:

Network edge:

- hosts (End systems):
  - Run application programs
  - e.g. Web, email
  - At "edge of network"
  - clients and servers
- Client/server model
  - Client host requests, receives service from always-on server
  - E.g. Web browser/server; email client/server
- Peer-to-peer model对等模式
  - Minimal (or no) use of dedicated servers
  - E.g. Skype, BitTorrent, KaZaA



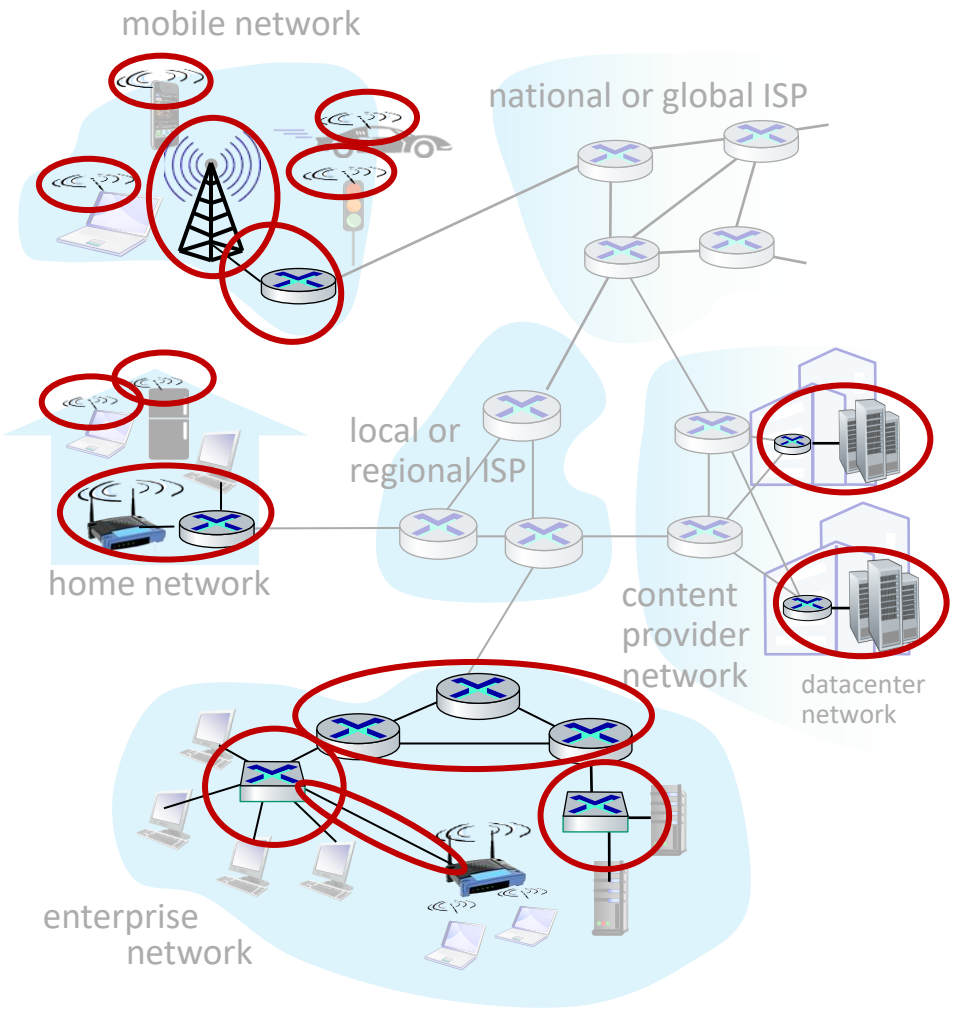
A closer look at network structure:

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at network structure:

Network edge:

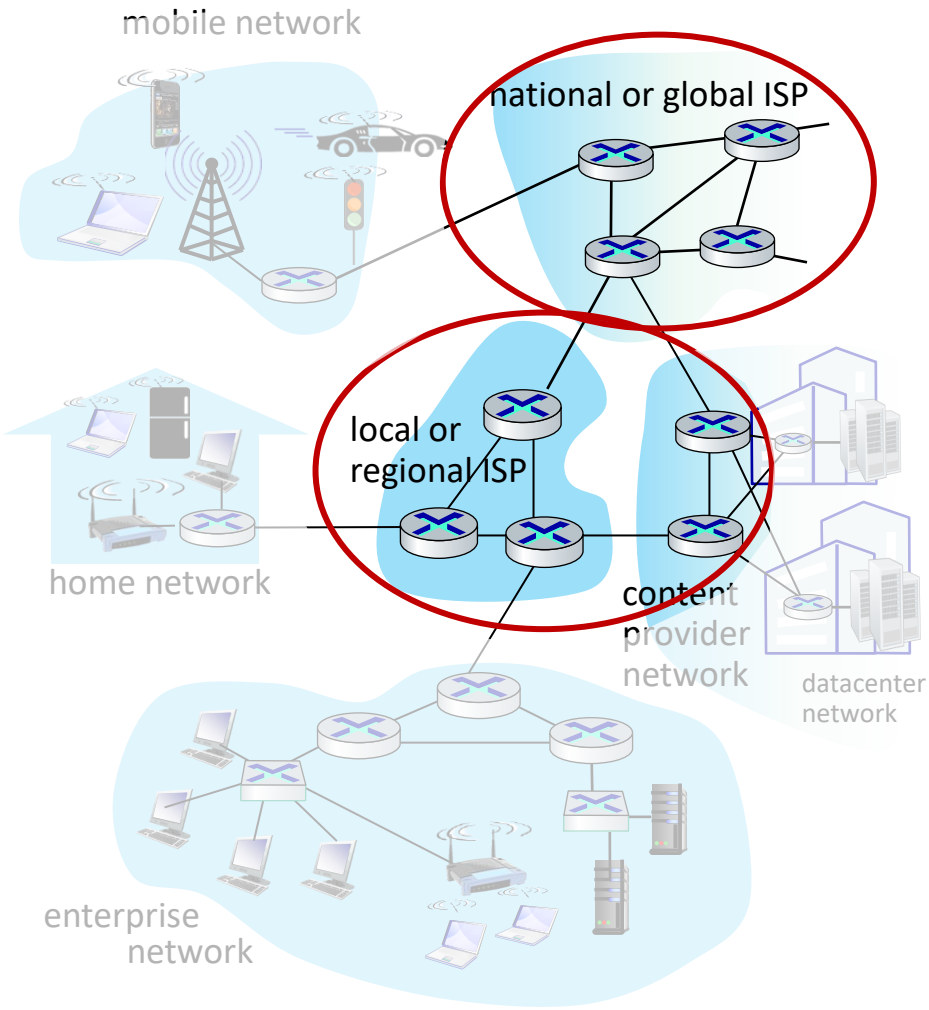
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links

Network core:

- interconnected routers
- network of networks



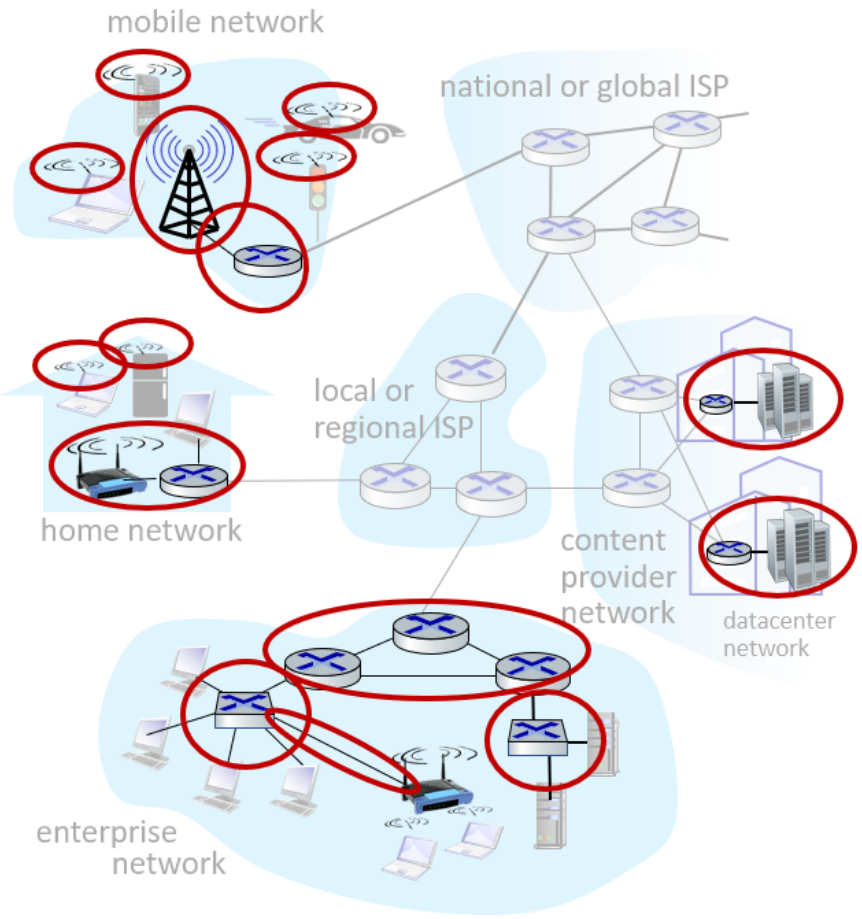
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

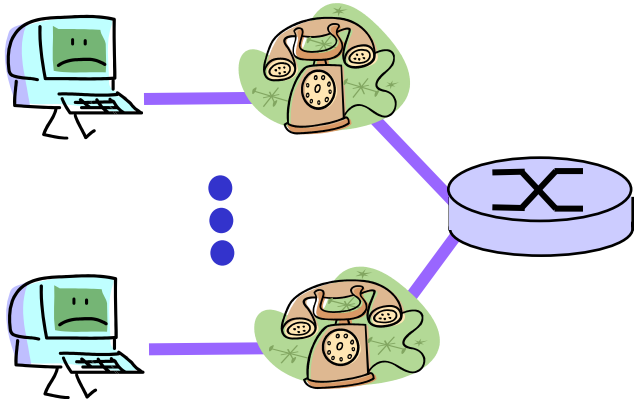
What to look for:

- bandwidth (bits per second) of access network?
- shared or dedicated?



Residential Access: Point to Point Access

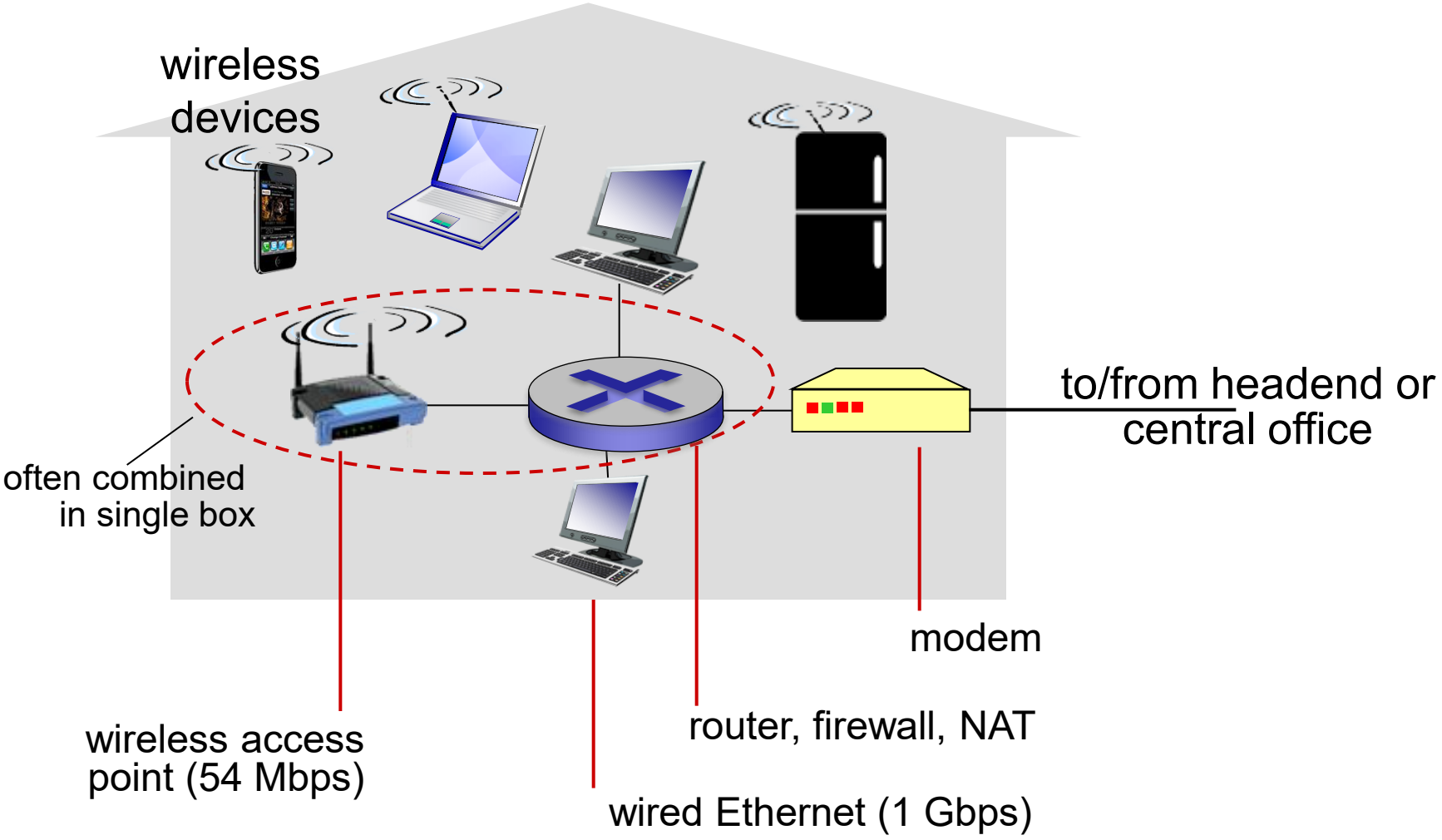
- Dialup via modem
  - Up to 56Kbps direct access to router (often less)
  - Can't surf and phone at same time: can't be "always on"



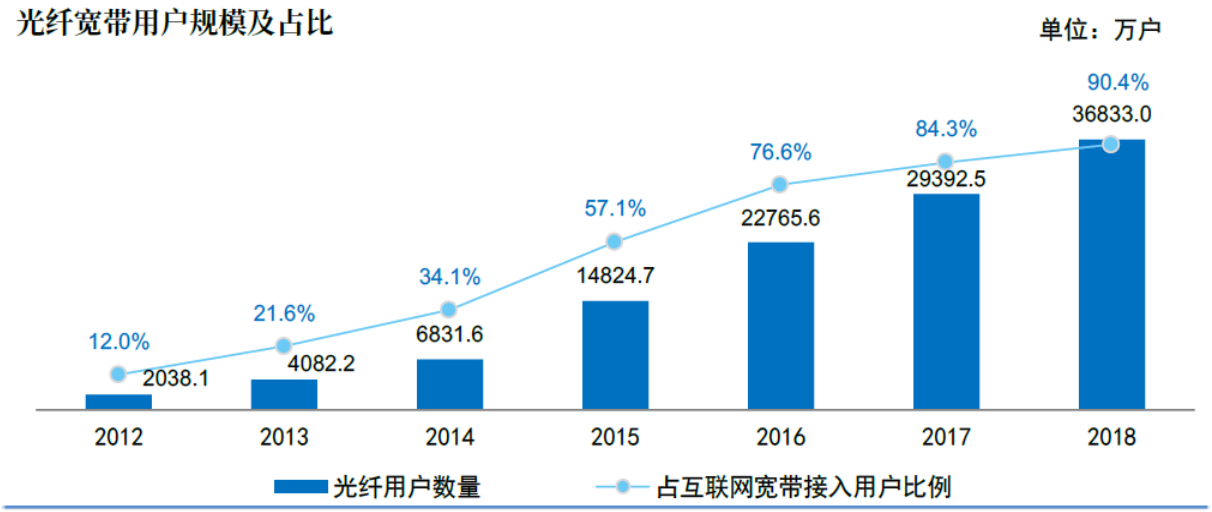
- ADSL: asymmetric digital subscriber line
  - Up to 1 Mbps upstream (today typically < 256 kbps)
  - Up to 8 Mbps downstream (today typically < 1 Mbps)

- HFC: hybrid fiber coax
  - Asymmetric: up to 30Mbps downstream, 2 Mbps upstream
  - Is shared broadcast medium

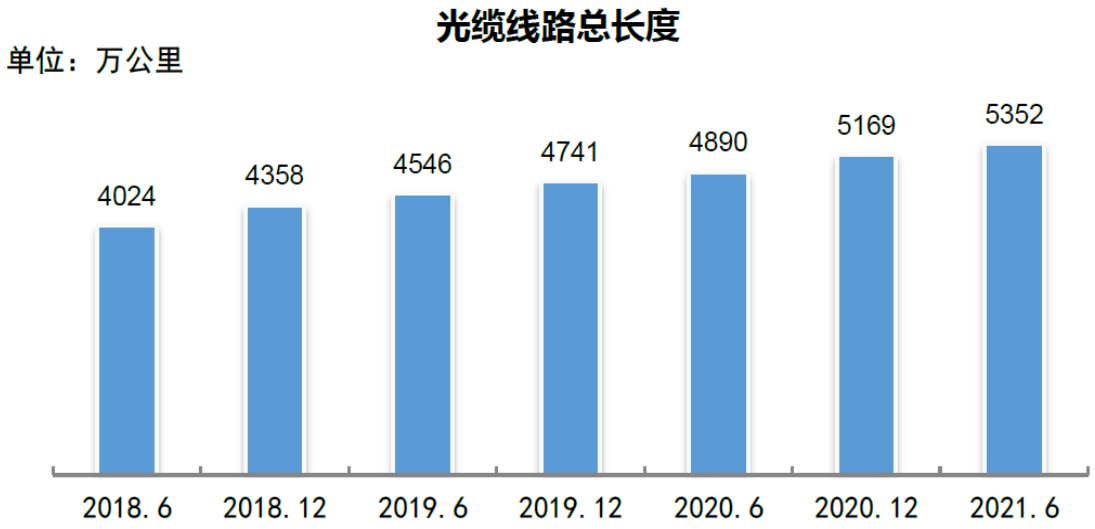
Access network: home network



Access network: home network

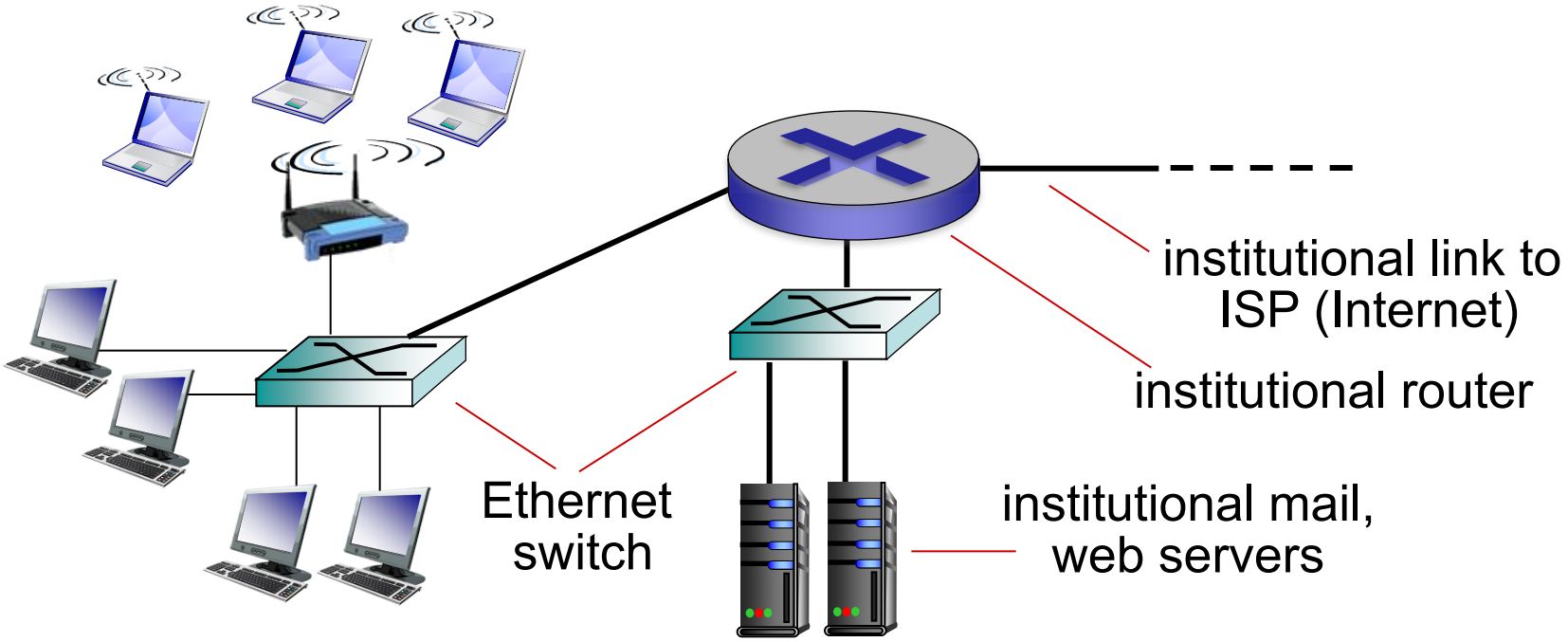


来源：工业和信息化部 2018.12



来源：工业和信息化部 2021.6

Enterprise access networks (Ethernet)



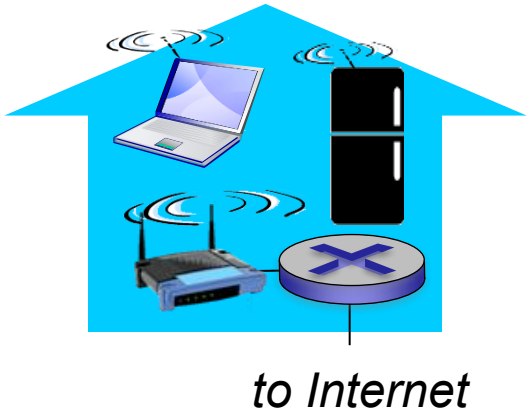
- typically used in companies, universities, etc.
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- today, end systems typically connect into Ethernet switch

Wireless access networks

- shared *wireless* access network connects end system to router
  - via base station aka "access point"

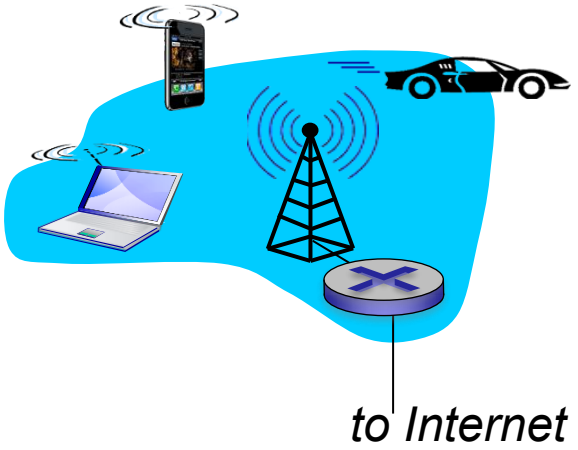
wireless LANs:

- within building (100 ft.)
- 802.11b/g/n/ac/ax (WiFi): 11, 54, 450 Mbps transmission rate



wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE, 5G

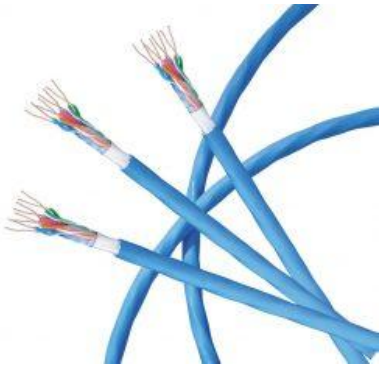


Physical media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

twisted pair (TP)

- two insulated copper wires两根绝缘铜线
  - Category 5: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10Gbps

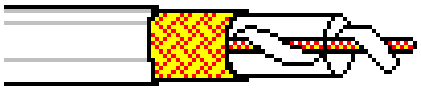




Physical media: coax, fiber

coaxial cable:同轴电缆

- two concentric copper conductors
- Bidirectional双向的
- broadband:
  - multiple channels on cable
  - HFC





fiber optic cable:光缆

- glass fiber carrying light pulses, each pulse a bit光脉冲
- high-speed operation:高速运行
  - high-speed point-to-point transmission (e.g., 10's-100's Gbps transmission rate)
- low error rate: 中继器间距大，不受电磁干扰
  - repeaters spaced far apart
  - immune to electromagnetic noise

中国光谷已成长为中国最大的光纤光缆、光电器件生产基地，最大的光通信技术研发基地。

Physical media: radio

- signal carried in electromagnetic spectrum电磁频谱中携带的信号
- no physical "wire"
- Bidirectional双向
- propagation environment effects:传播环境的影响
  - reflection
  - obstruction by objects物体阻挡
  - Interference干涉

radio link types:

- terrestrial microwave陆地微波
  - e.g. up to 45 Mbps channels
- LAN (e.g., WiFi)局域网
  - 54 Mbps
- wide-area (e.g., cellular)
  - 4G cellular: ~ 10 Mbps
- Satellite卫星
  - Kbps to 45Mbps channel (or multiple smaller channels)
  - 270 msec end-end delay
  - geosynchronous versus low altitude

Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching分组交换, circuit switching电路交换, network structure

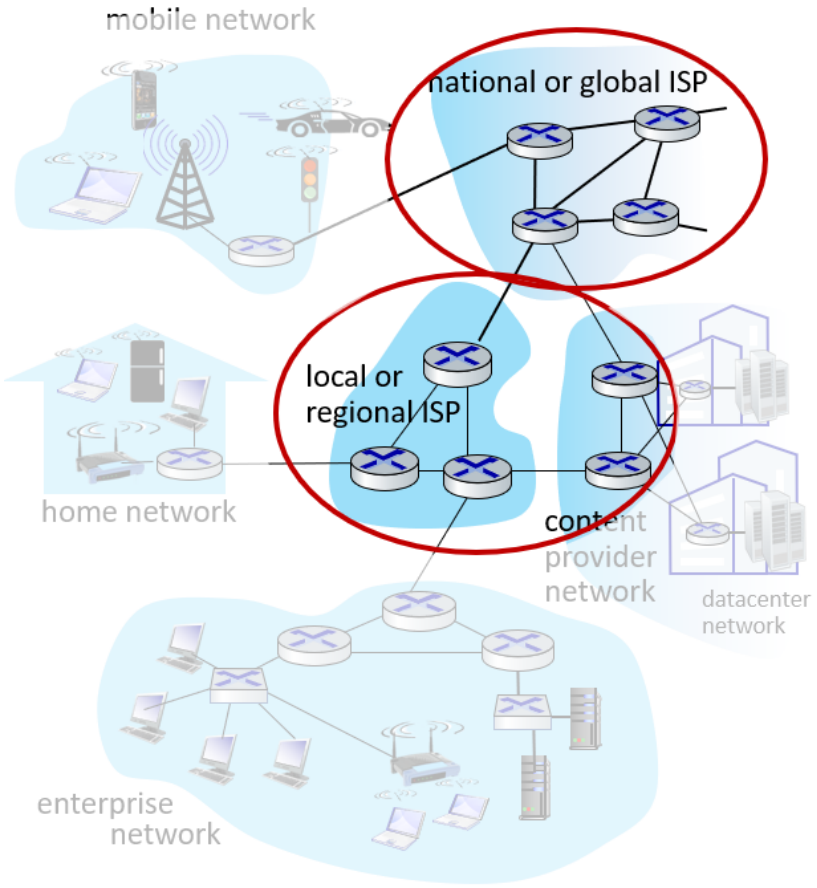
1.4 Delay & loss in packet-switched networks

1.5 Protocol layers, Service models

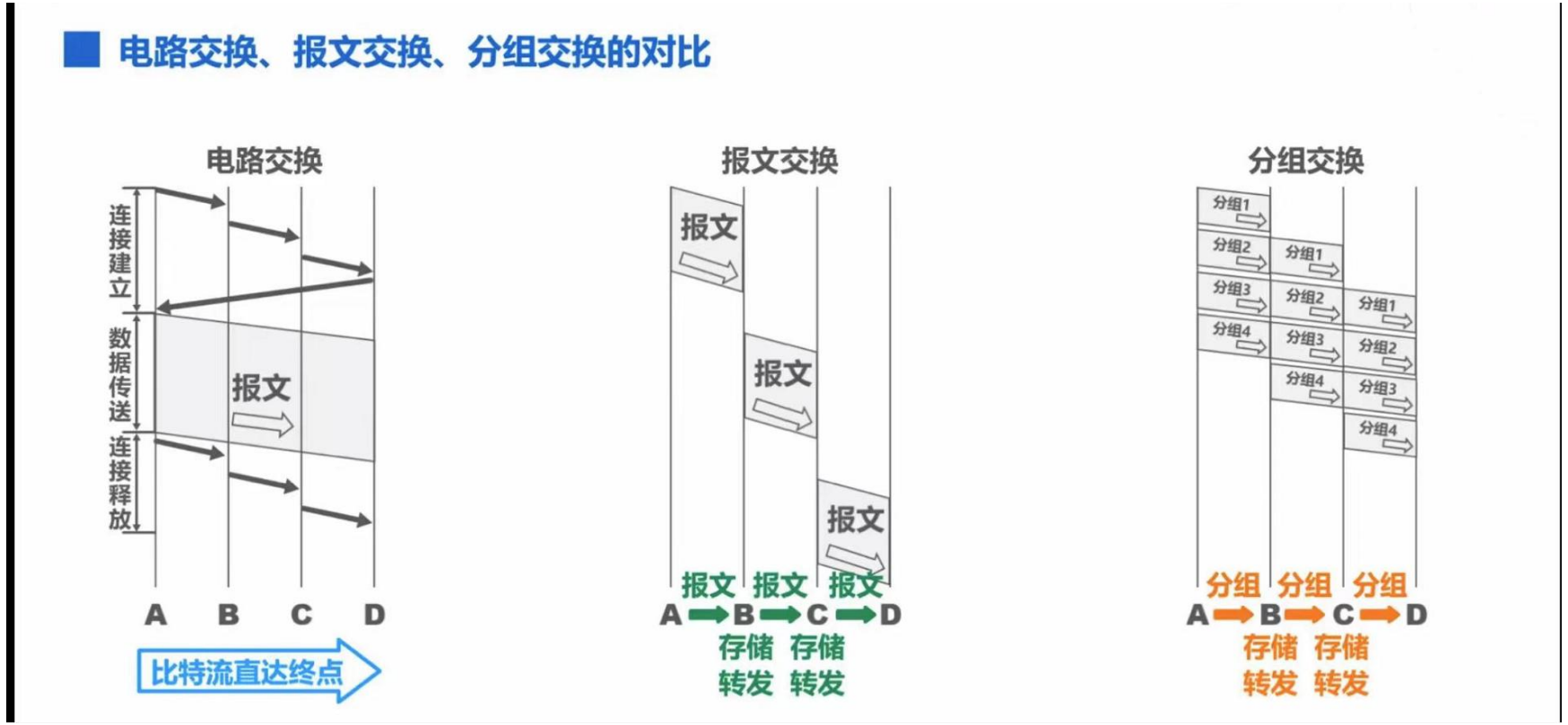
1.6* Internet history

The network core

- mesh of interconnected routers互联网路由器
- The fundamental根本的 question: how is data transferred through net 数据是如何通过网络转送的?
  - Circuit-switching: dedicated circuit per call: telephone net
  - Packet-switching: data sent through net in



The network core

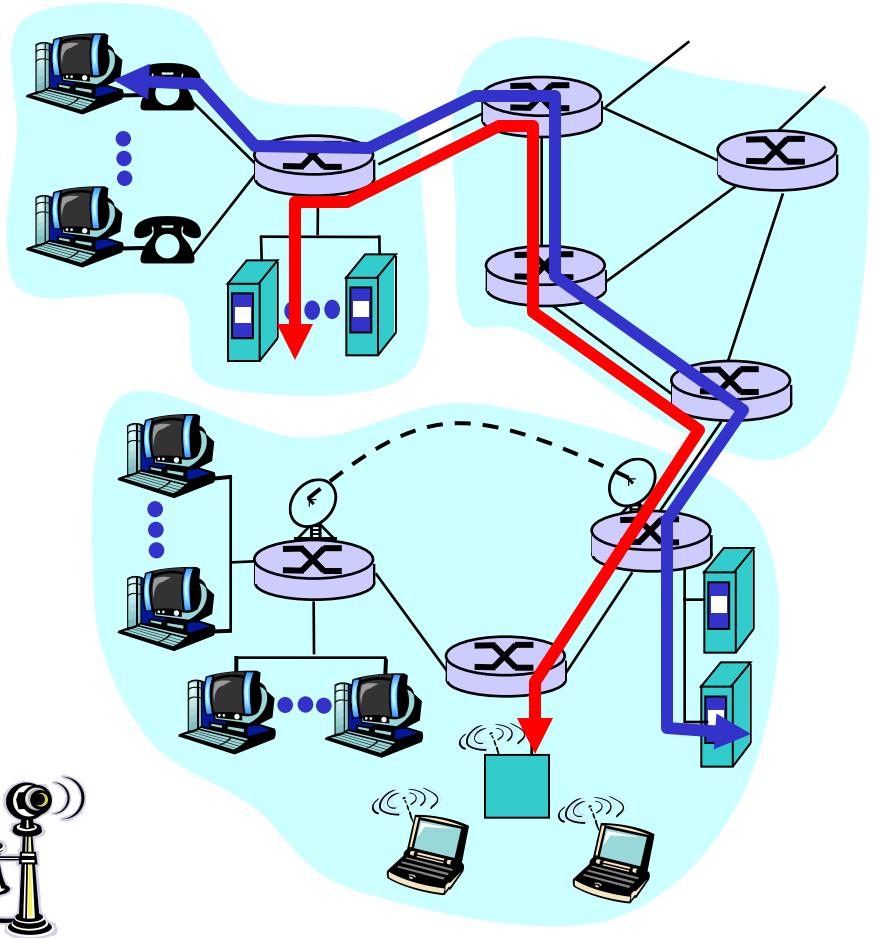


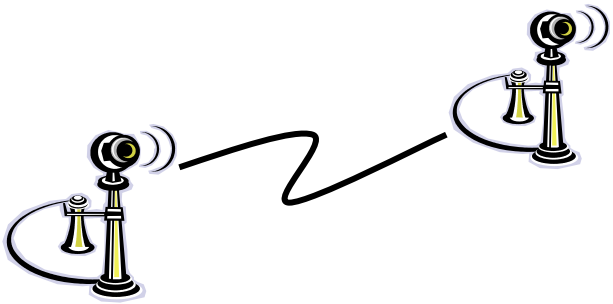
Message switching and Packet-Switching: Store-and-Forward存储转发

Circuit Switching

End-end resources reserved for "call"预留资源

- Link bandwidth链路带宽, switch capacity交换容量
- Dedicated resources: no sharing专用资源，不共享
- Circuit-like (guaranteed) performance
- Call setup required





Circuit Switching

Network resources (e.g., bandwidth) divided into "pieces" 资源分块

- Pieces allocated to calls 分配给通话的pieces
- Resource piece *idle* if not used by owning call *(no sharing)*



- Dividing link bandwidth into "pieces"
  - Frequency division频分
  - Time division时分



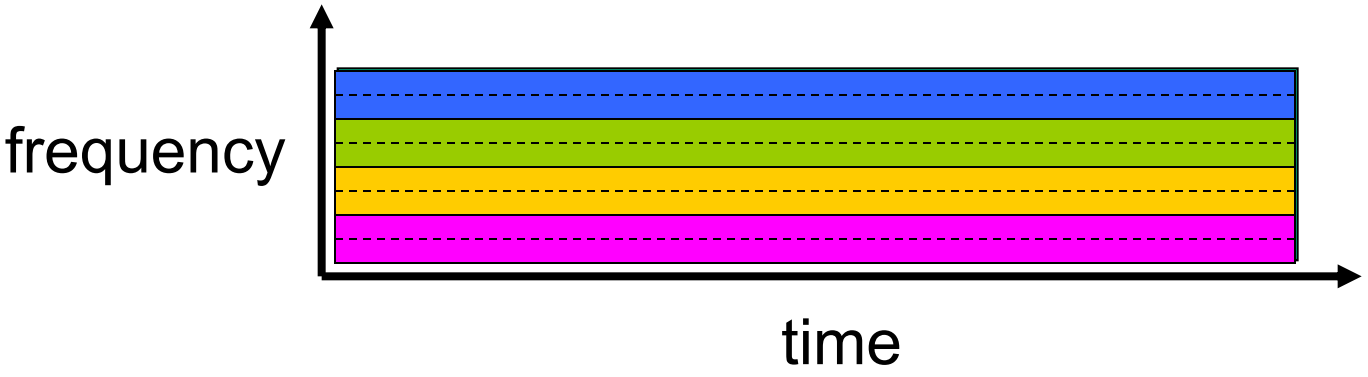
Circuit switching: FDM versus TDM

Example:

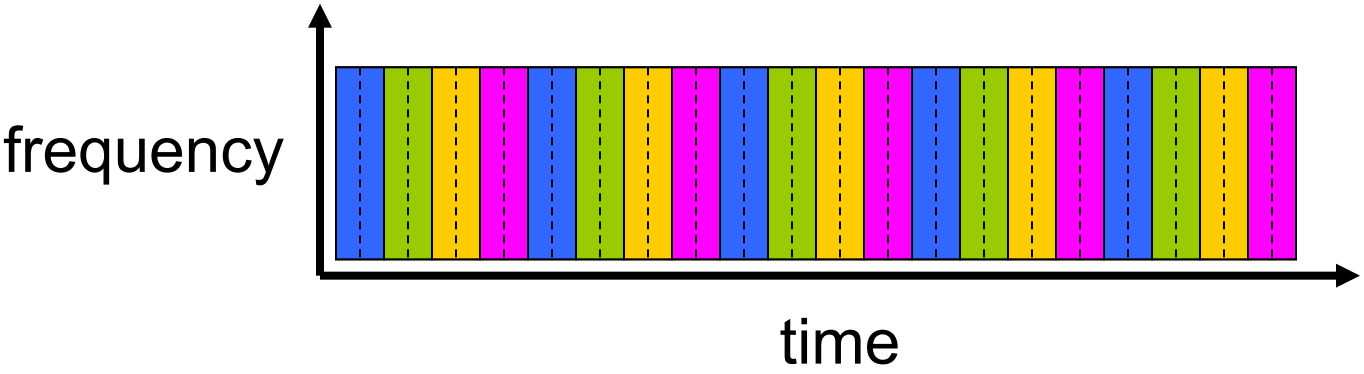
4 users



FDM



TDM



Numerical Example

- How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
  - All links are 1.536 Mbps
  - Each link uses TDM with 24 slots
  - 500 msec to establish end-to-end circuit
  - 教材P20

Work it out!



Network Core: Packet Switching

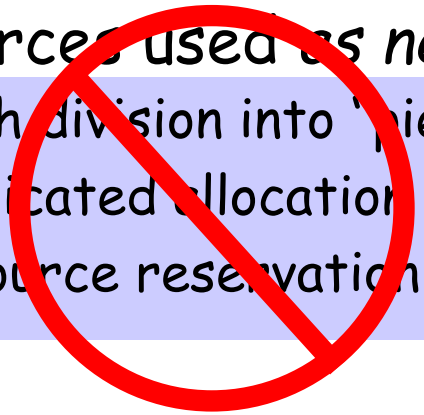
Each end-end data stream divided into *packets*

- User A, B packets *share* network resources
- Each packet uses full link bandwidth
- Resources used *as needed*

Bandwidth division into "pieces"

Dedicated allocation

Resource reservation

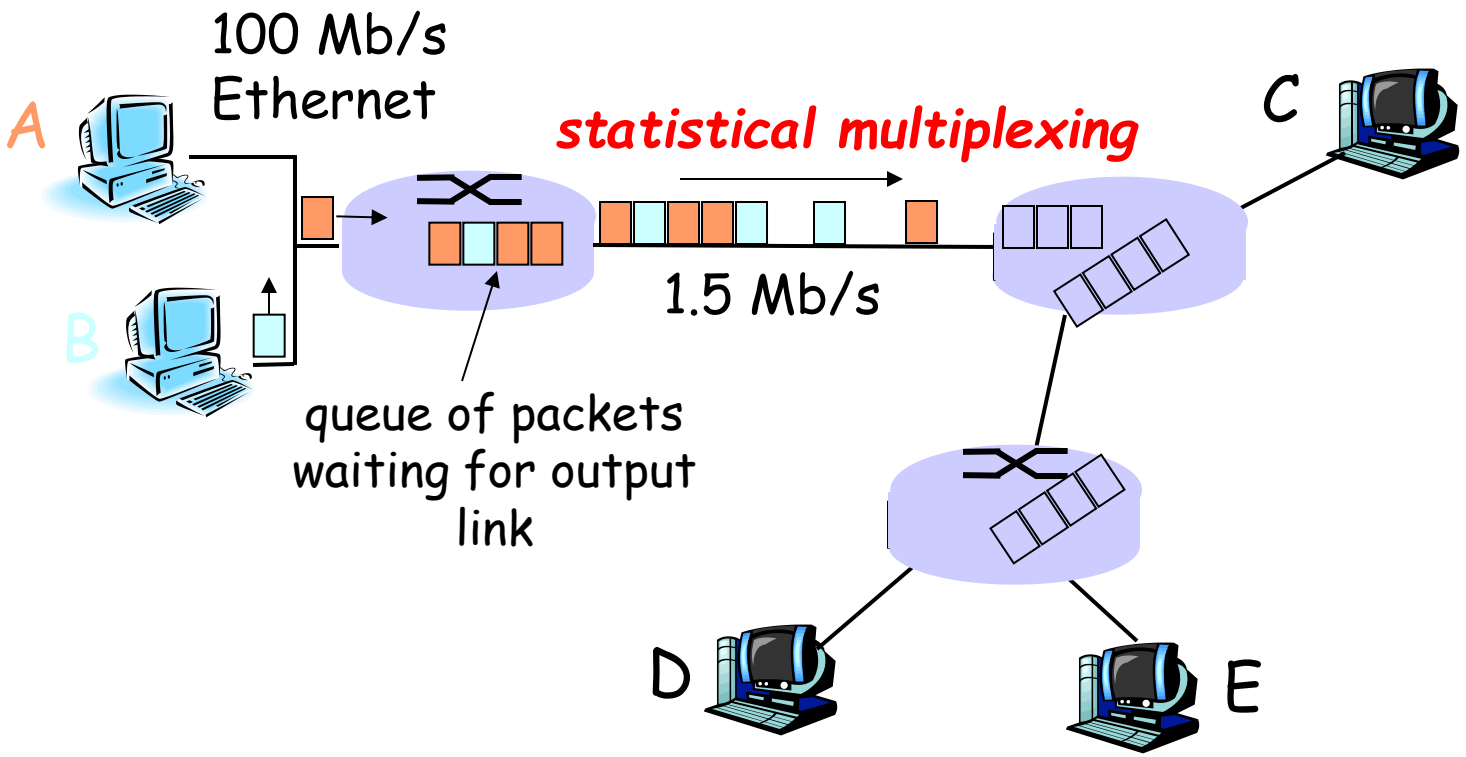


节点收到完整数据包后再转发

Resource contention:

- Aggregate resource demand can exceed amount available资源需求总量可能超过可用数量
- Congestion: packets queue, wait for link use拥塞：数据包排队，等待链接
- Store and forward: packets move one hop at a time一跳一次
  - Node receives complete packet before forwarding

Packet Switching: Statistical Multiplexing



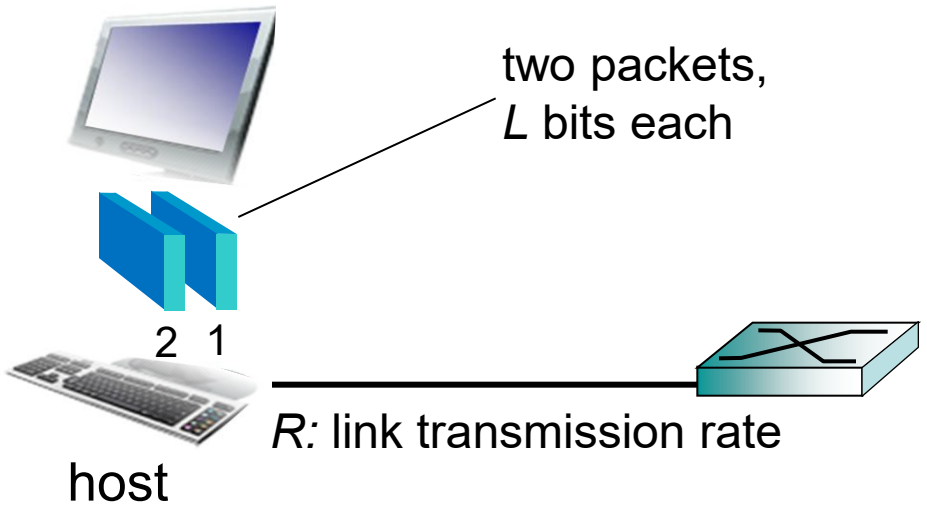
Sequence of A & B packets does not have fixed pattern, shared on demand ***statistical multiplexing***

TDM: each host gets same slot in revolving TDM frame

Host: sends packets of data

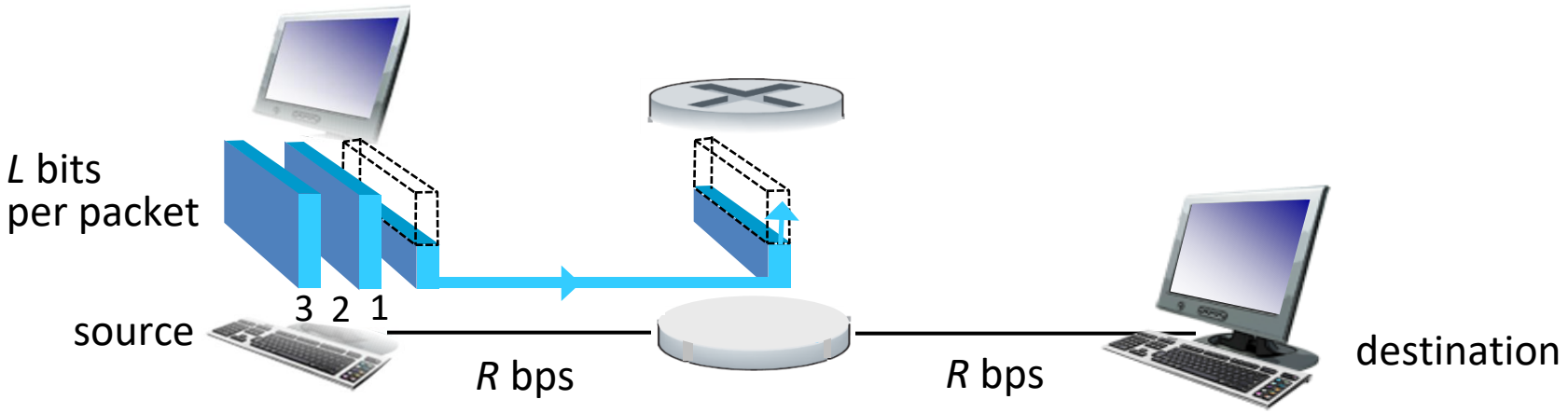
host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Packet-switching: store-and-forward



- Transmission delay: takes *L*/*R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- store and forward : entire packet must arrive at router before it can be transmitted on next link
- end-end delay = 2*L*/*R* (assuming zero propagation delay)从一个端系统到另一个端系统

one-hop numerical example:

- *L* = 7.5 Mbits
- *R* = 1.5 Mbps
- one-hop transmission delay = 5 sec

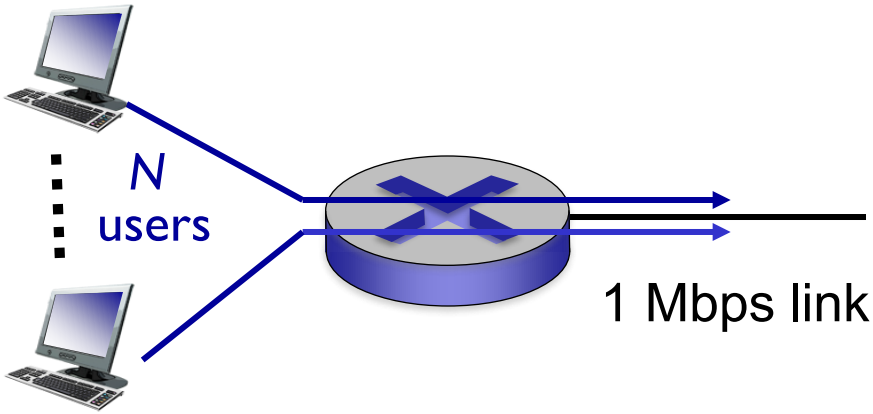
more on delay shortly …

Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time



- *circuit-switching:*
  - 10 users
- *packet switching:*
  - with 35 users, probability > 10 active at same time is less than .0004 *

Q: how did we get value 0.0004?

Q: what happens if > 35 users ?

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

Packet switching versus circuit switching

- Probability that a user is active p=0.1
- Probability that a user is inactive q=1-p
- We want to find N such that concurrent users are more than 10

$$\Pr\{Aggregate\,input > 1Mbps\} < 0.001$$

$$\Pr\{k\ out\ of\ N\ are\ active\ and\ k > 10\} < 0.001$$

$$\sum_{i=11}^{N} \binom{N}{i} p^i (1-p)^{N-i} < 0.001$$

Packet switching versus circuit switching

is packet switching a "slam dunk winner?"

- great for bursty data
  - resource sharing
  - simpler, no call setup
- excessive congestion possible: packet delay and loss
  - protocols needed for reliable data transfer, congestion control
- Q: How to provide circuit-like behavior?
  - bandwidth guarantees needed for audio/video apps
  - still an unsolved problem (chapter 7)

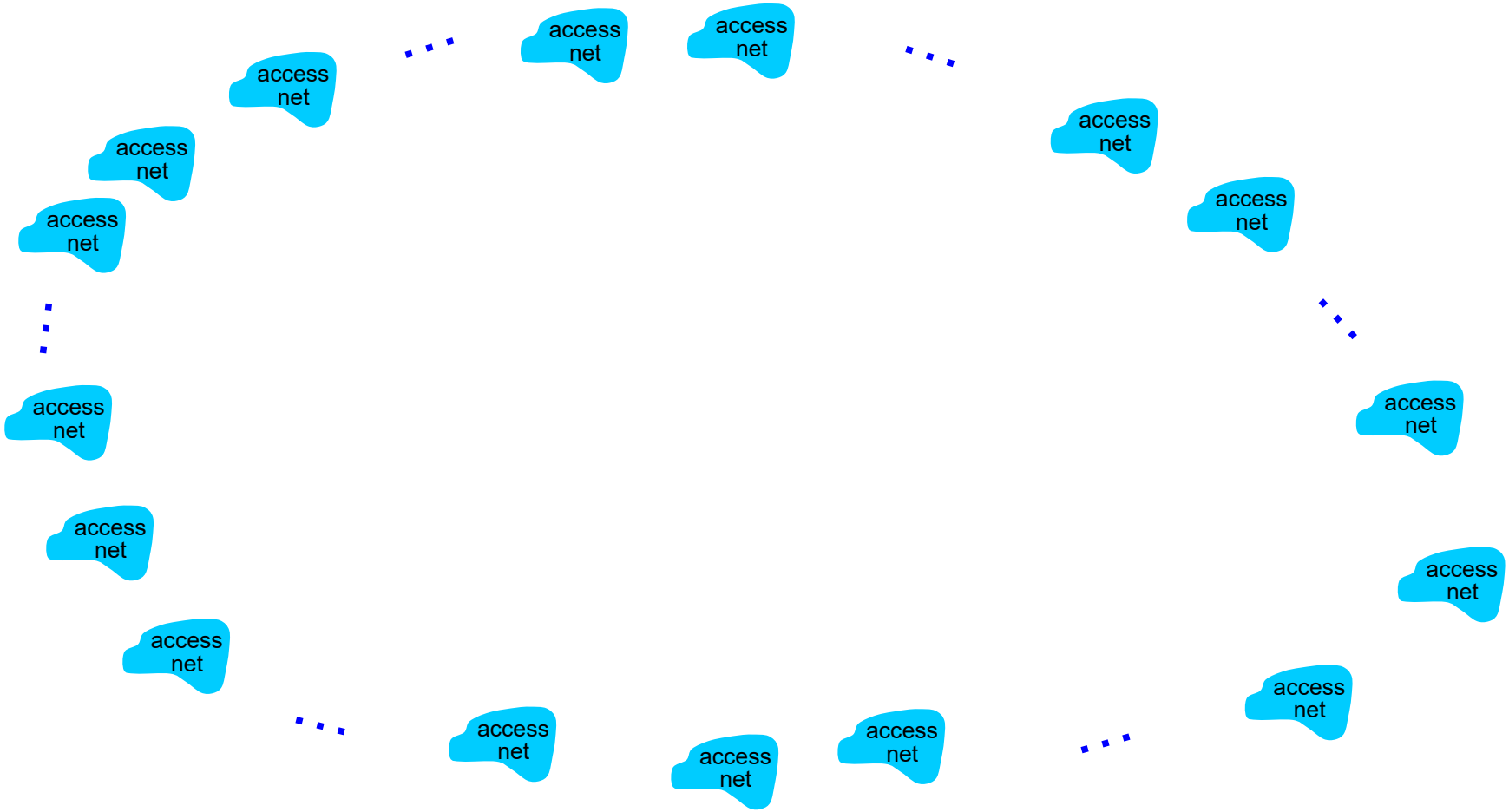
Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?

Internet structure: network of networks

- End systems connect to Internet via access ISPs (**Internet Service Providers**)
  - residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - so that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

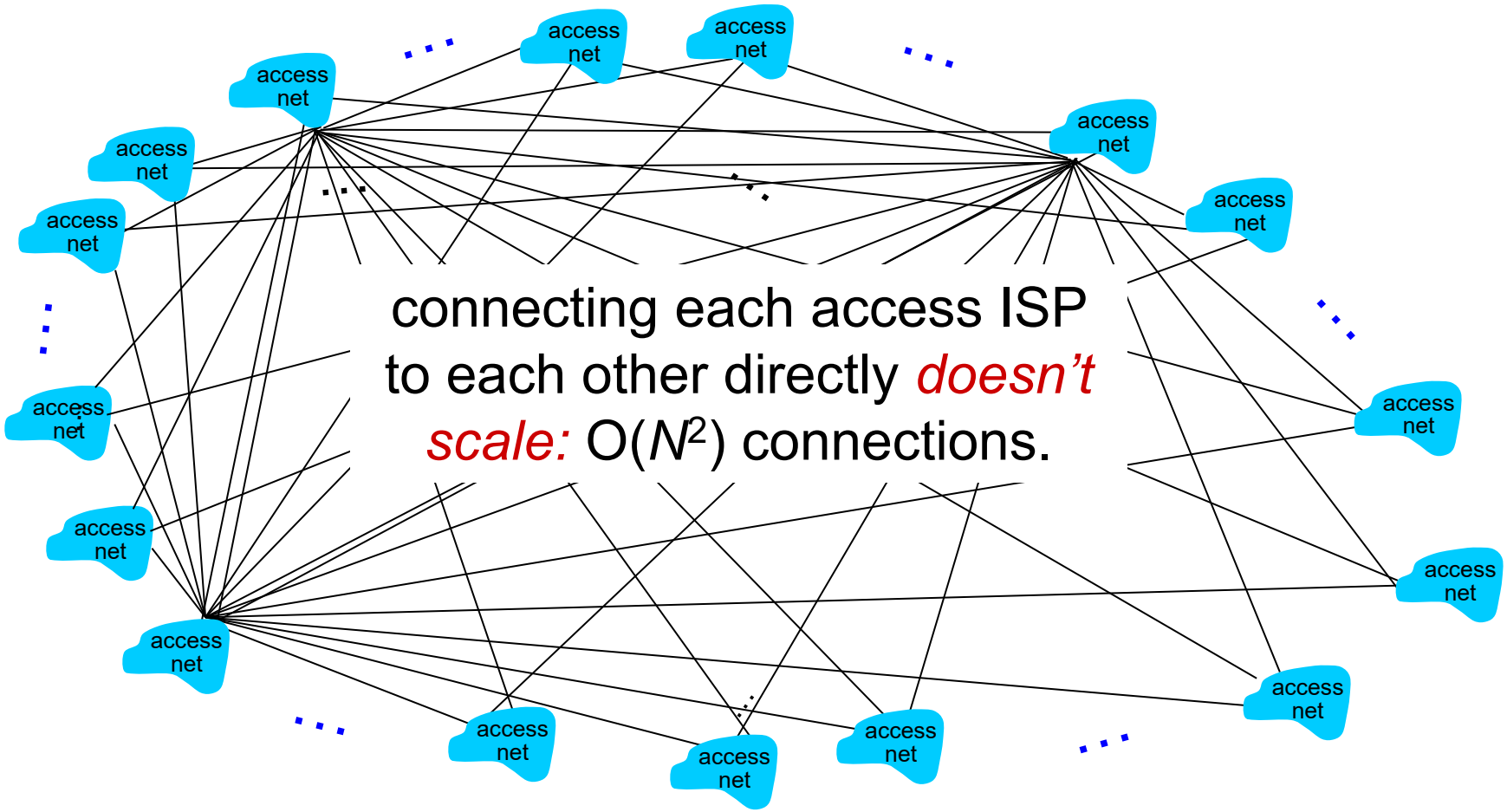
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: network of networks

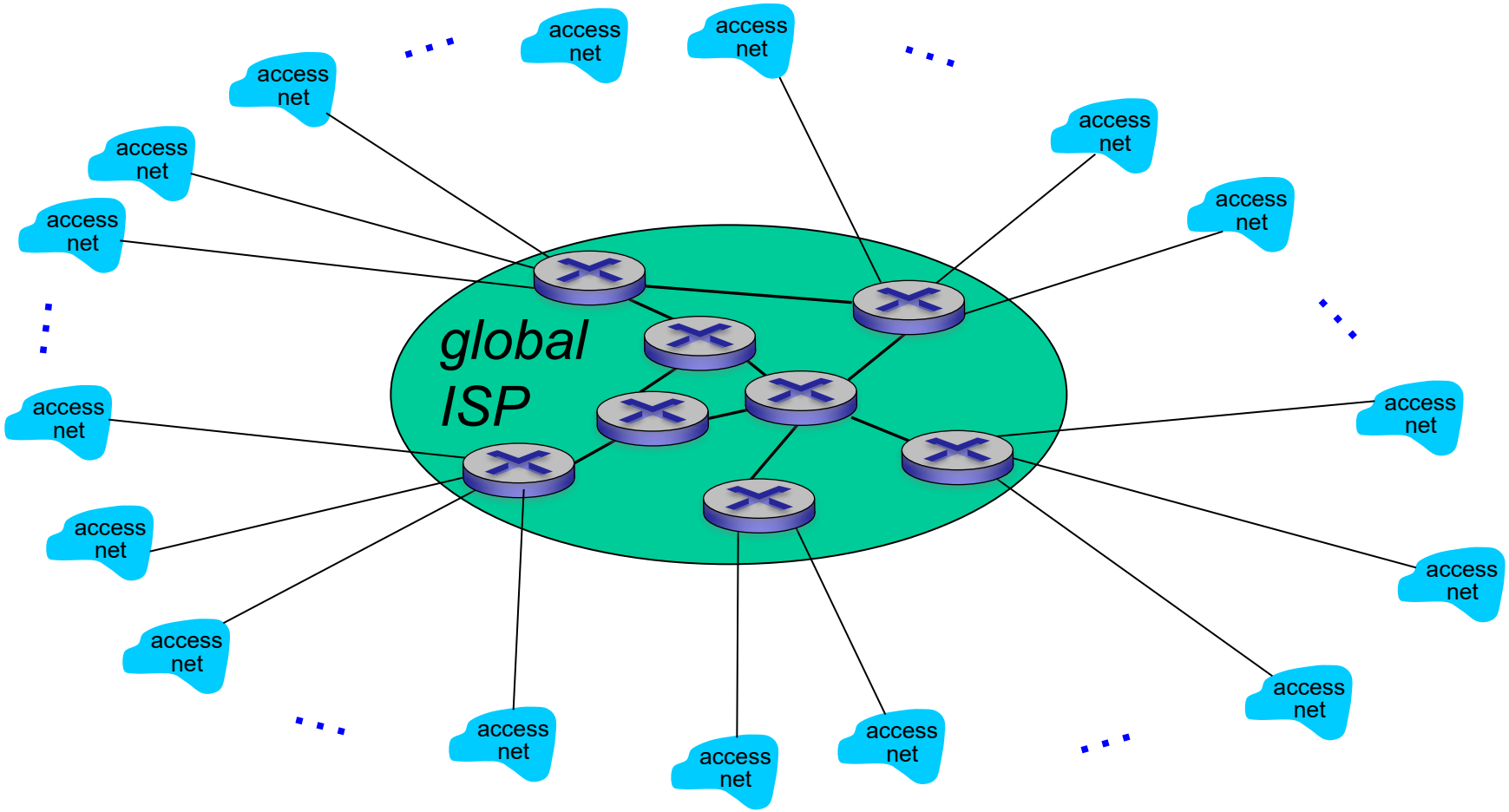
Option: *connect each access ISP to every other access ISP?*



Internet structure: network of networks

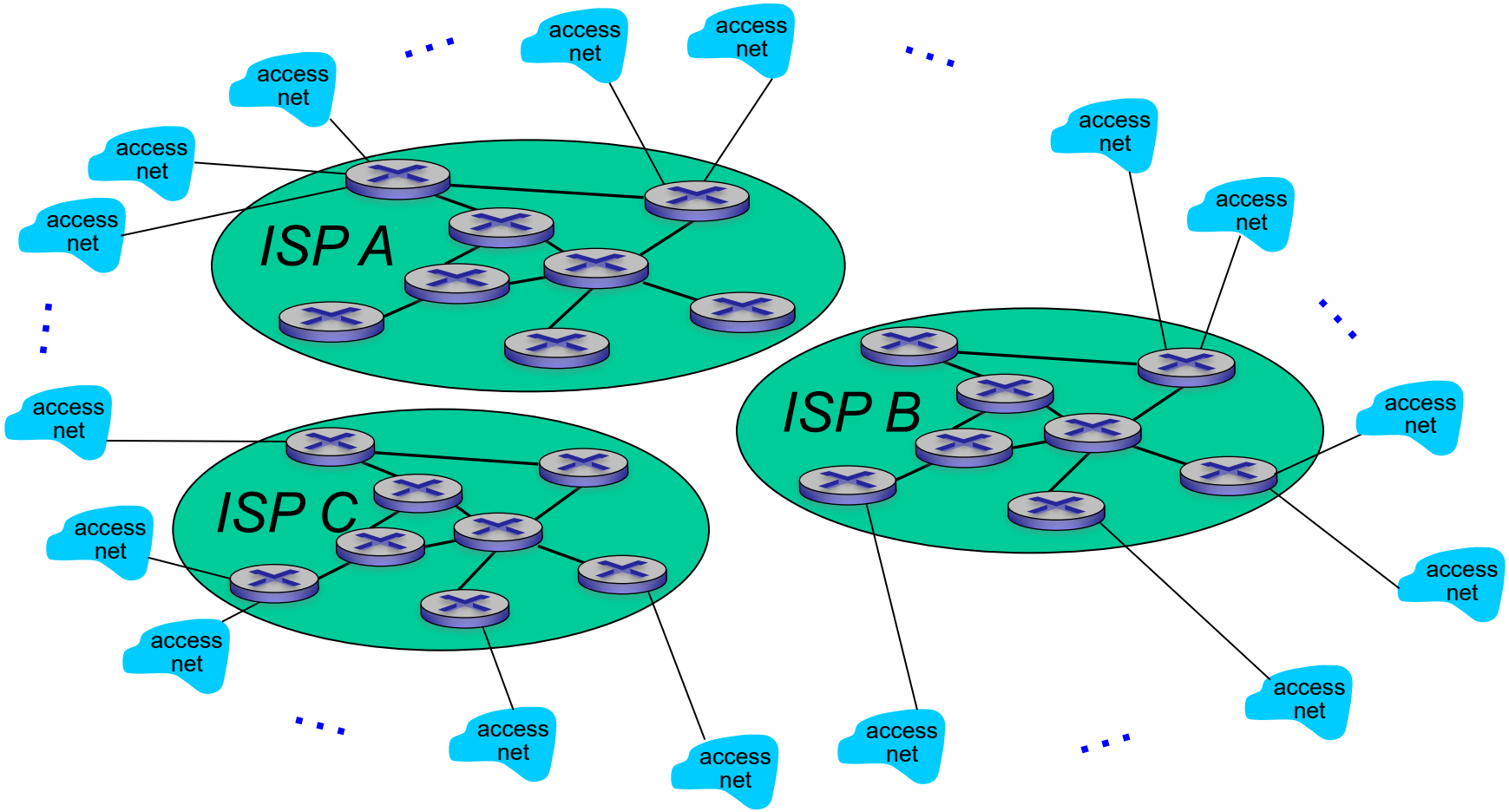
Option: connect each access ISP to one global transit ISP?

Customer and *provider* ISPs have economic agreement.



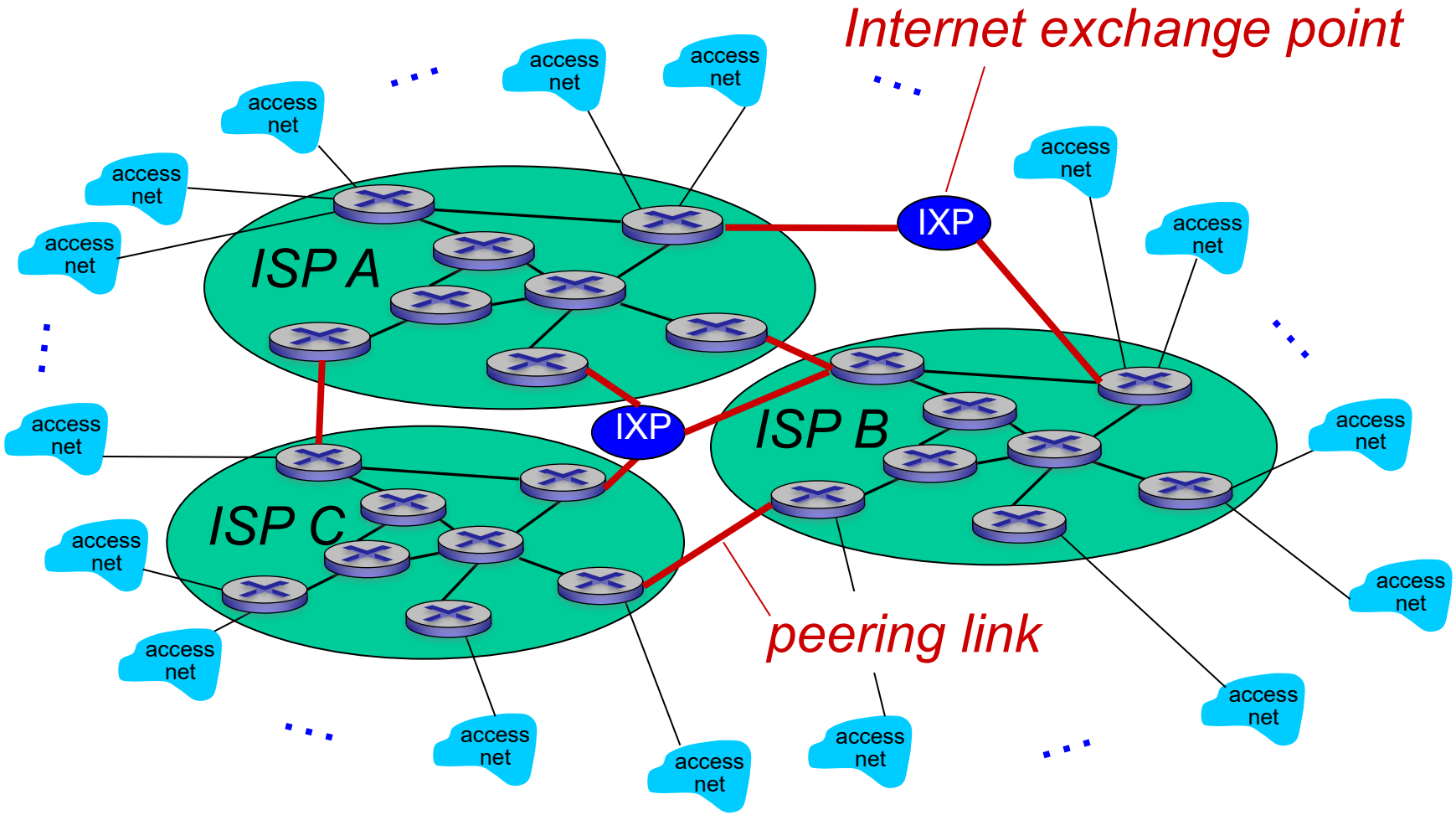
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors ….



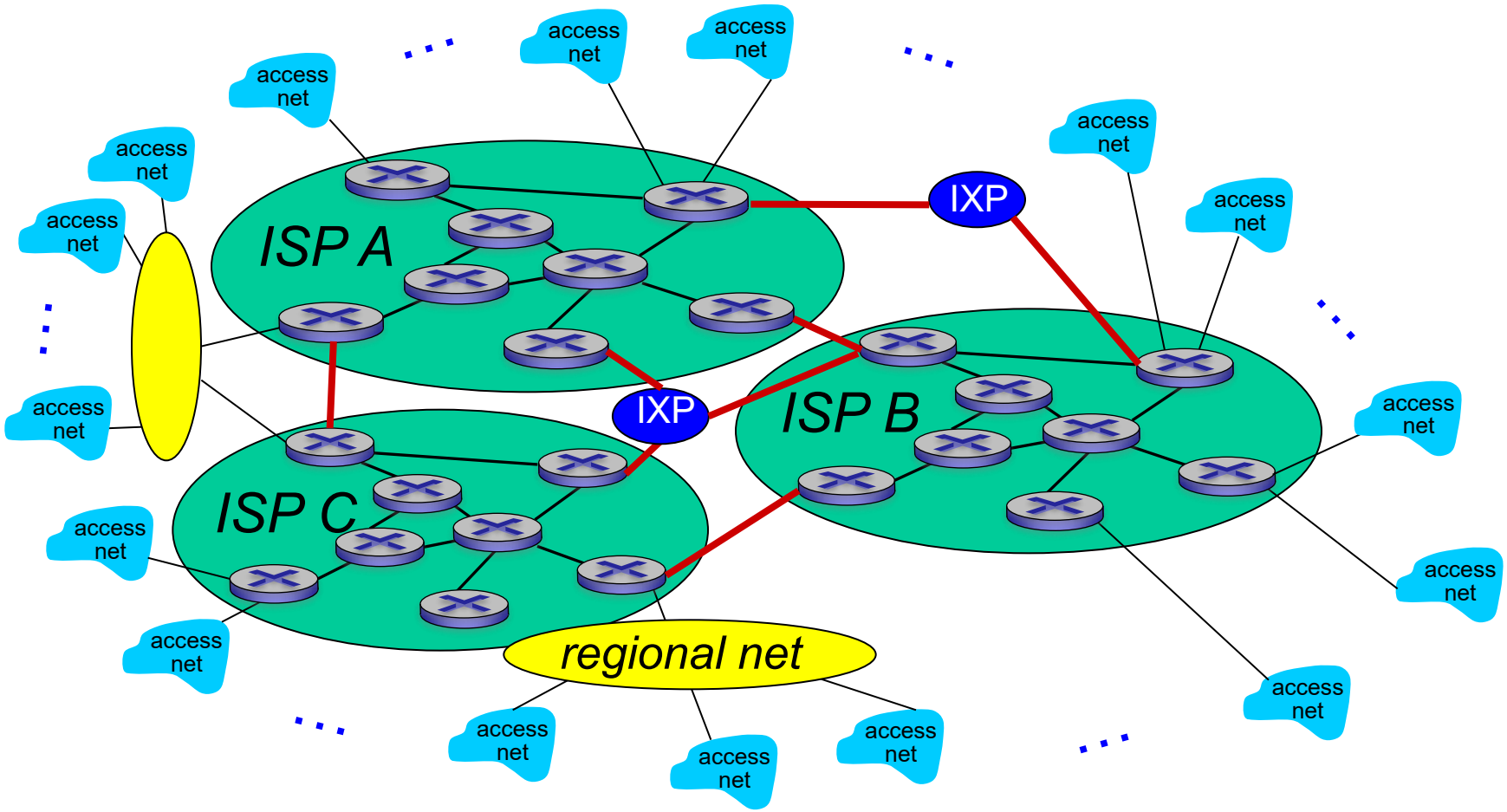
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors …. which must be interconnected



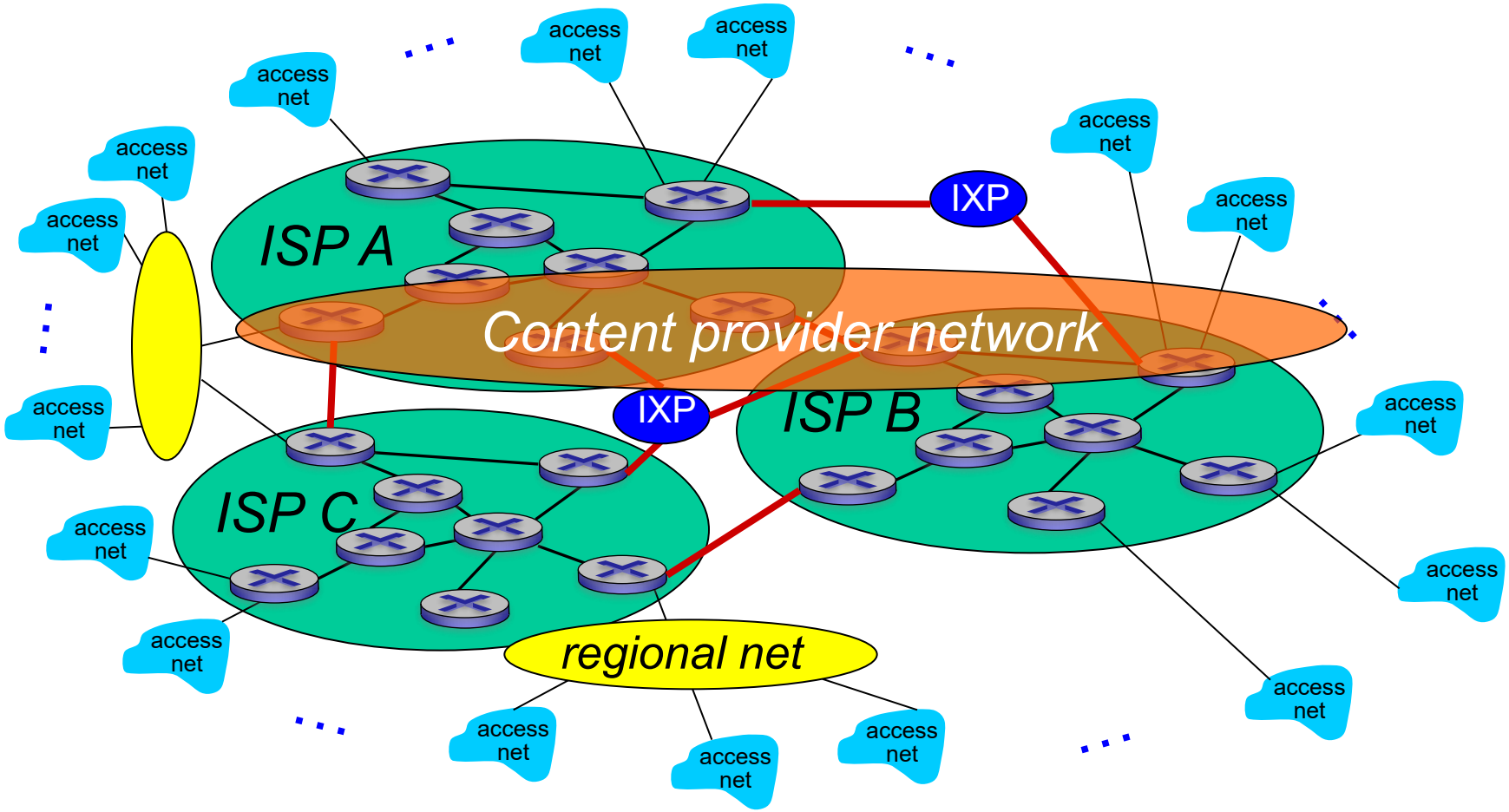
Internet structure: network of networks

… and regional networks may arise to connect access nets to ISPs

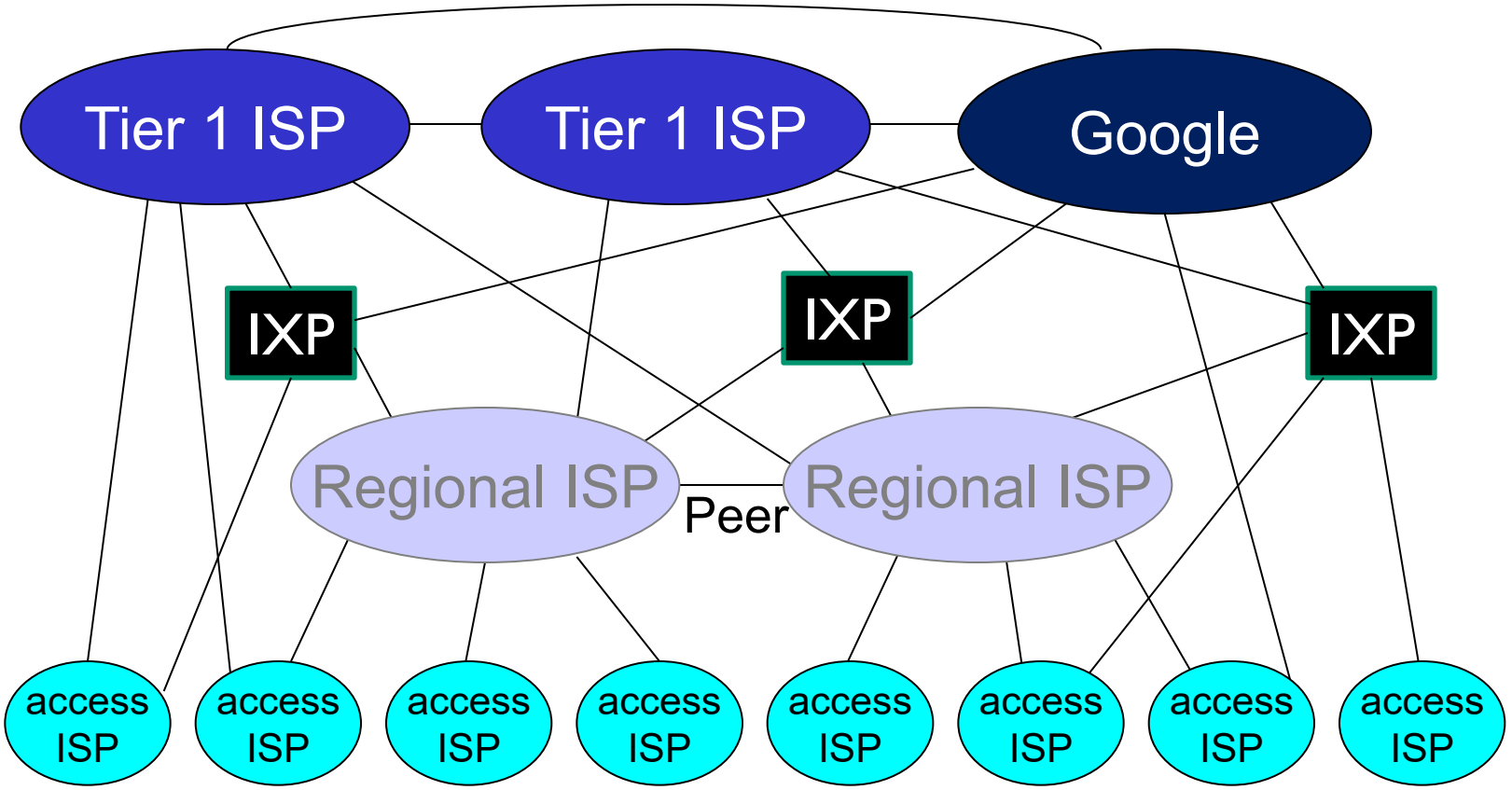


Internet structure: network of networks

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users

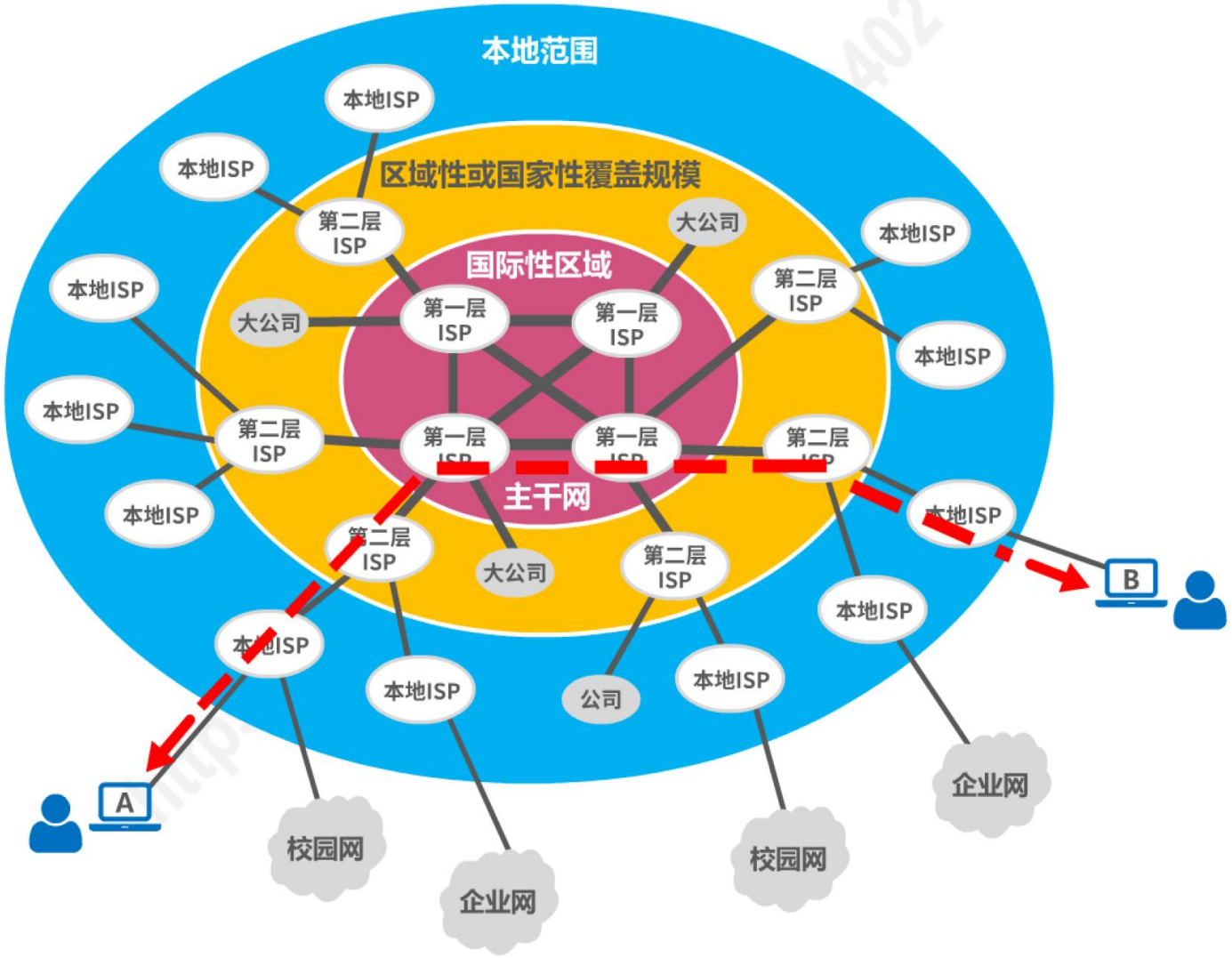


Internet structure: network of networks



- access ISPs, regional ISPs, tier-1 ISPs,
- PoPs, multi-homing, peering, and IXPs

Internet structure: network of networks



Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

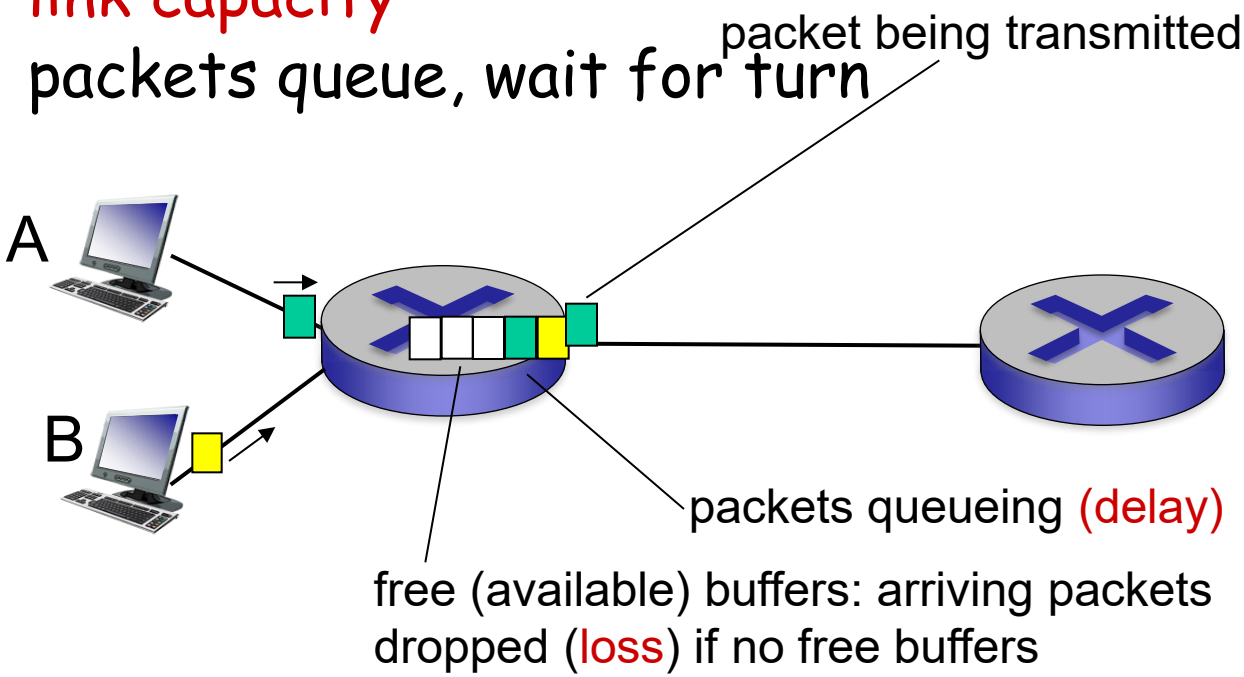
1.5 Protocol layers, Service models

1.6* Internet history

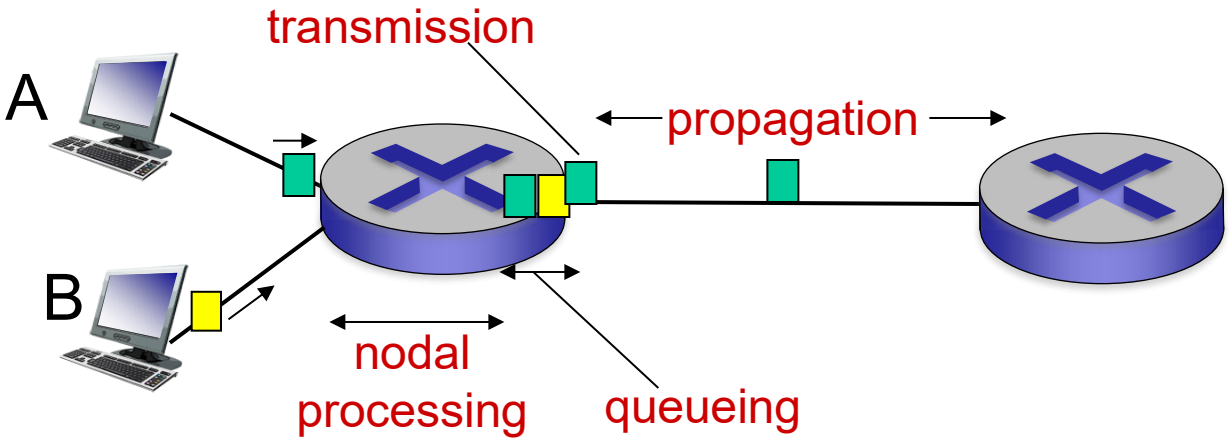
How do loss and delay occur?

packets *queue* in router buffers路由器缓冲区中的数据包队列

- packet arrival rate to link (temporarily) exceeds output link capacity
- packets queue, wait for turn



Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

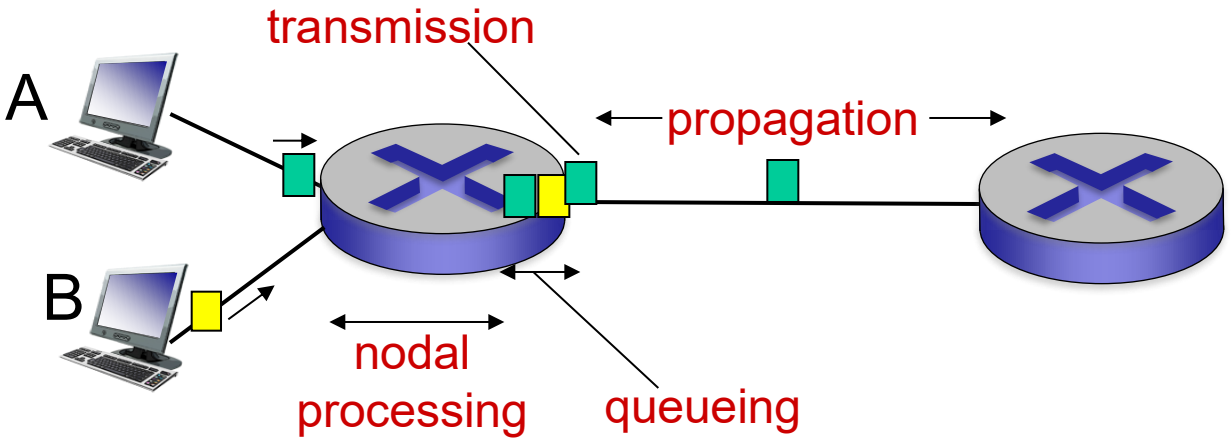
d_{proc}: nodal processing 处理时延

- check bit errors查错
- determine output link
- typically < msec

d_{queue}: queueing delay排队时延

- time waiting at output link for transmission
- depends on congestion level of router路由器的拥塞程度

Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay传输时延:

- L: packet length (bits)
- R: link ~~bandwidth~~ (bps)
- $d_{trans} = L/R$

d_{trans} and d_{prop} *very* different

d_{prop}: propagation delay传播时延:

- d: length of physical link
- s: propagation speed (~2×10^8 m/sec)
- $d_{prop} = d/s$

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/
* Check out the Java applet for an interactive animation on trans vs. prop delay

Excersice-1(nodal delay)

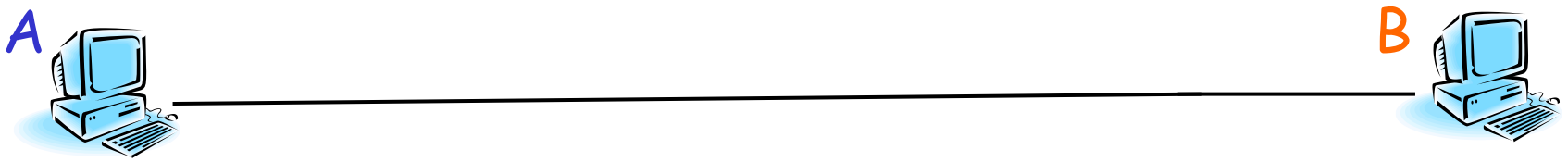
Assuming that the link length *d* between two hosts is 60m, the link bandwidth **R** is 10Mbps, and the signal propagation rate *s* is 2×10^8 m / s. Now host A sends 1-bit packet to host B.

1) How long does it take for host B to receive this packet? If host A starts transmitting the packet at time $t = 0$, how far does the front end of the signal of the packet spread at time $t = d_{trans}$?

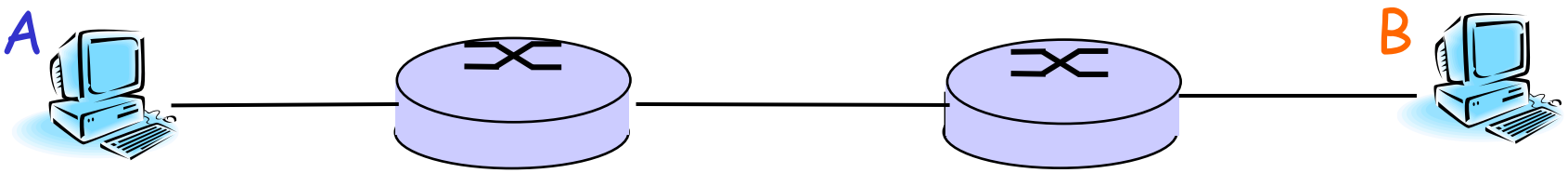
2) What if the link length *d* is 20 meters?

3) If the link length *d* is 10 meters, what is the specific transmission situation?

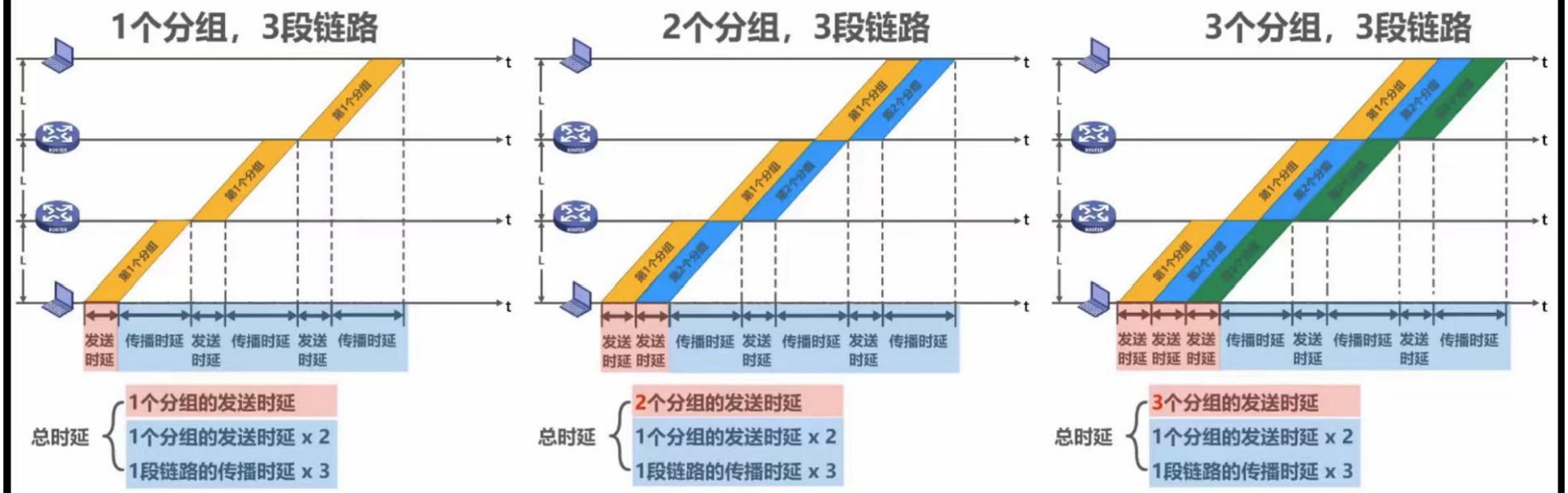
4) If host A sends n-bit packet to host B, how long does it take for host B to receive the packet?



End-to-end delay

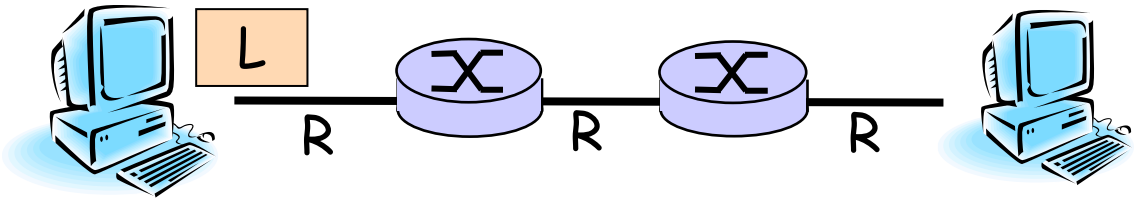


假设：分组等长，各链路长度相同、带宽也相同，忽略路由器的处理时延



若n个分组，m段链路，则总时延是多少？

Exercise-2 (end-to-end delay)



- Message switching and Packet switching: Entire message or packet must arrive at router before it can be transmitted on next link: *store and forward*
- Calculate the end-to-end delay of message switching and packet switching

Example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- t_{prop} : propagation delay of each link
- Divide the message into 5000 small packets: l=1.5 kbits

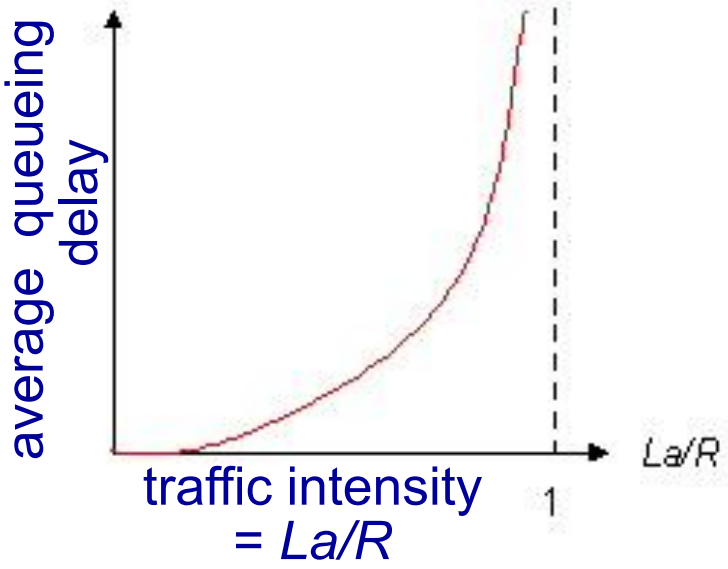
Exercise-3 (end-to-end delay)

The message to be sent is *x* (bits) in total. There are *k* links from the source to the destination. The propagation delay of each link is *d*(s) and the bandwidth is *R*(bps). In circuit switching, the circuit establishment time is *s*(s). In packet switching, the message can be divided into several packets with *p*(bits) length. It is assumed that the processing time and queuing time of the message/packets at each node are ignored.

Q: under what conditions is the end-to-end delay of circuit switching greater than that of packet switching?

Queueing delay (revisited)

- *R*: link bandwidth (bps)
- *L*: packet length (bits)
- a: average packet arrival rate



- *La/R* ~ 0: avg. queueing delay small
- *La/R* -> 1: avg. queueing delay large
- *La/R* > 1: more "work" arriving than can be serviced, average delay infinite!



La/R ~ 0

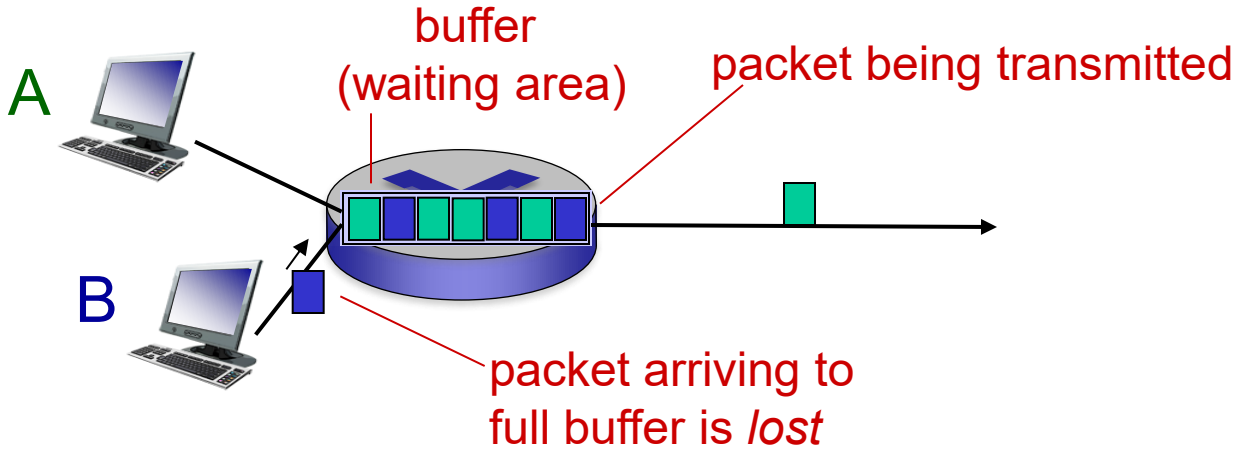


La/R -> 1

* Check online interactive animation on queuing and loss

Packet loss

- queue (aka buffer) preceding link in buffer has finite capacity缓冲区容量有限
- when packet arrives to full queue, packet is dropped (aka lost)队列已满，丢弃分组
- lost packet may be retransmitted by previous node, by source end system, or not at all可能重传



* Check out the Java applet for an interactive animation on queuing and loss

Chapter 1: roadmap

1.1 What is the Internet?

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- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

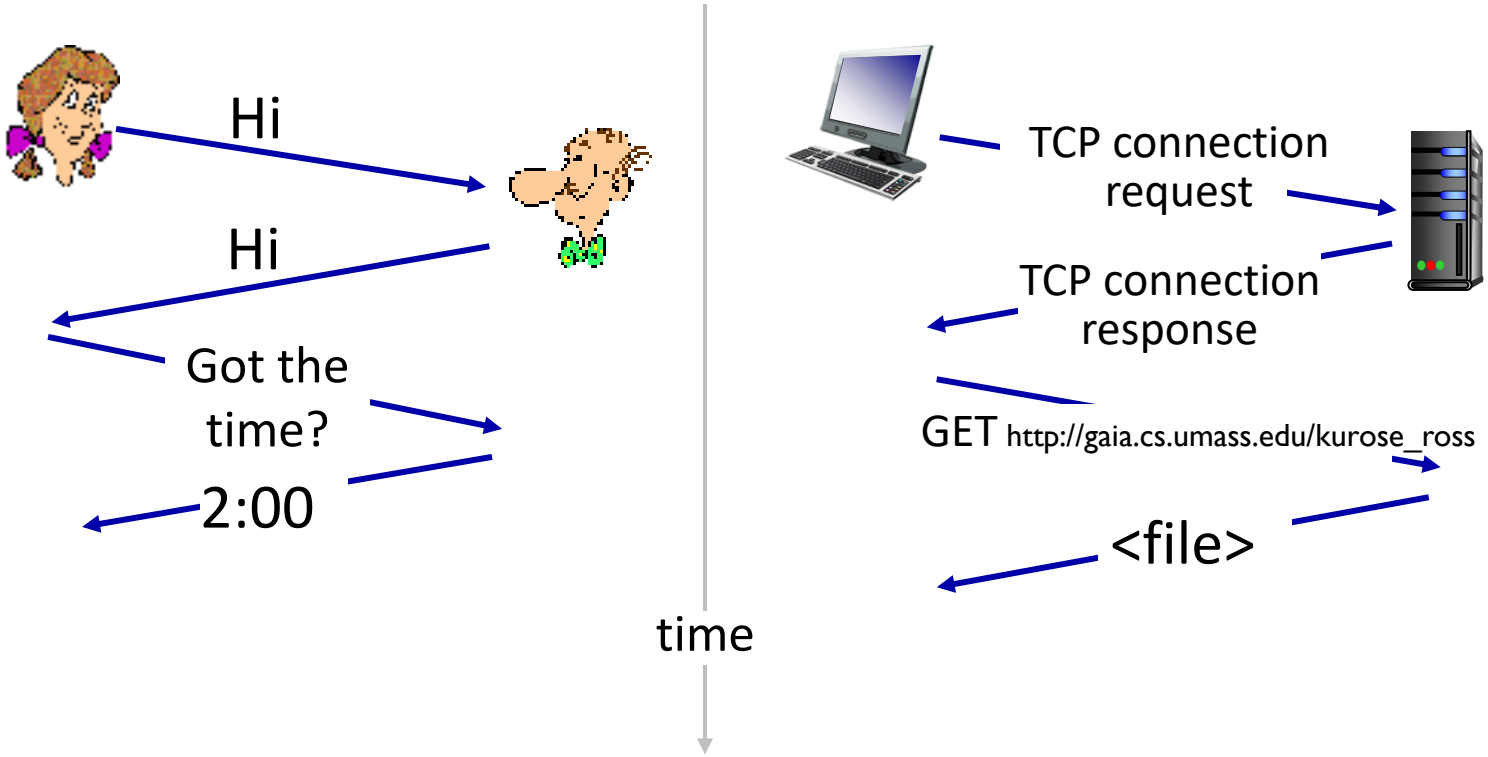
1.4 Delay & loss in packet-switched networks

1.5 Protocol layers, Service models

1.6* Internet history

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

What's a protocol?

Human protocols:

- "what's the time?"
- "I have a question"
- introductions

Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols互联网上所有通信活动受协议约束

Rules for:一些规则举例

… specific messages sent

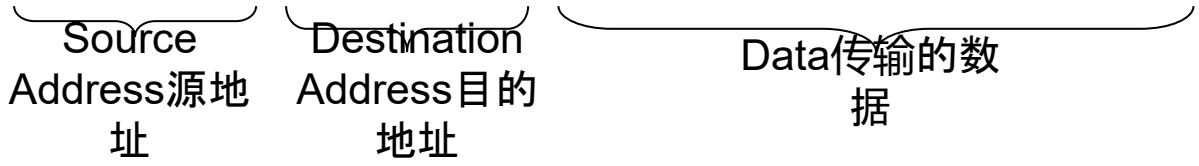
… specific actions taken when message received, or other events

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt 协议定义了格式。信息发送接受顺序，在传输信息时执行的操作

3 elements of a protocol

- Syntax: Defines how to do it定义操作方法

Example: 10101101101110110100110011110…



- Semantics: Defines what it is语义:定义是什么
- Timing: Defines the order and speed定义时序和速度

Protocol "layers"

Networks are complex, with many "pieces":

- Hosts主机
- Routers路由器
- links of various media不同媒介的链路
- Applications应用
- Protocols协议
- hardware, software硬件软件

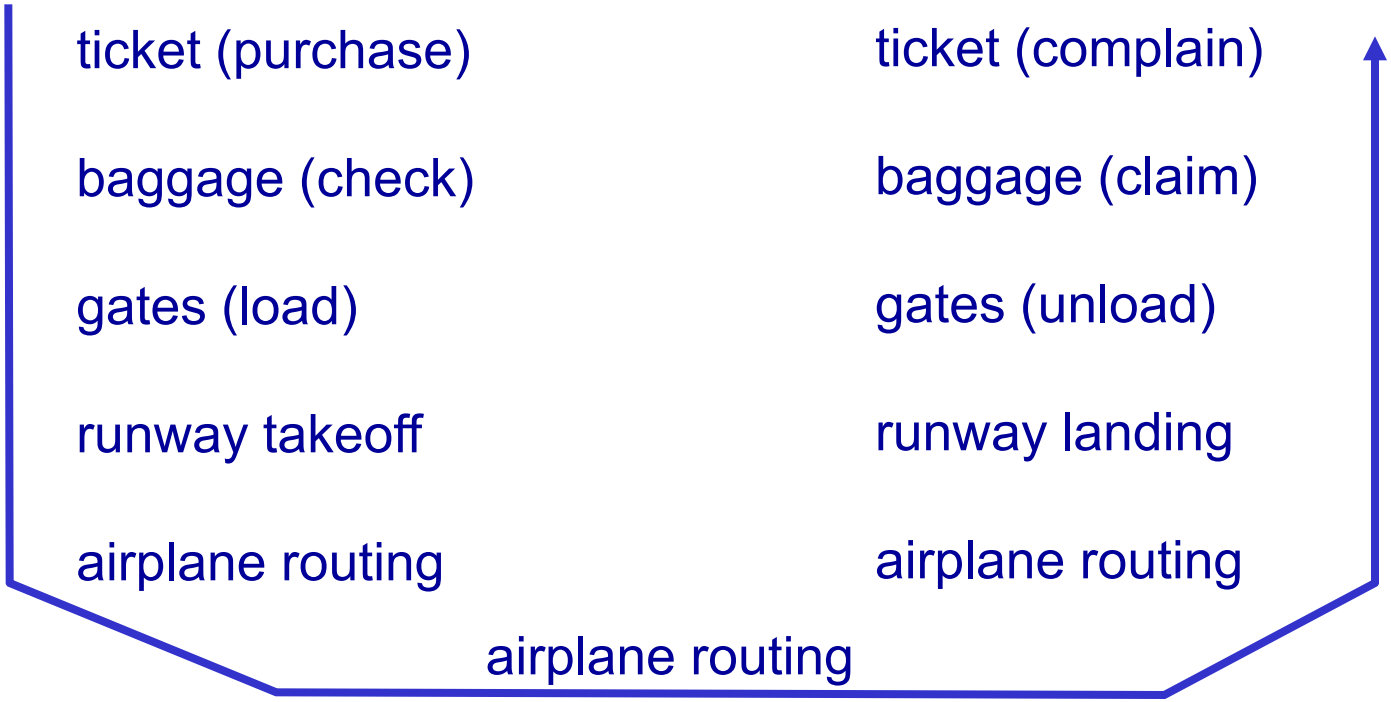
Question:

is there any hope of *organizing* structure of network?

是否有希望描绘/讨论网络的结构层次

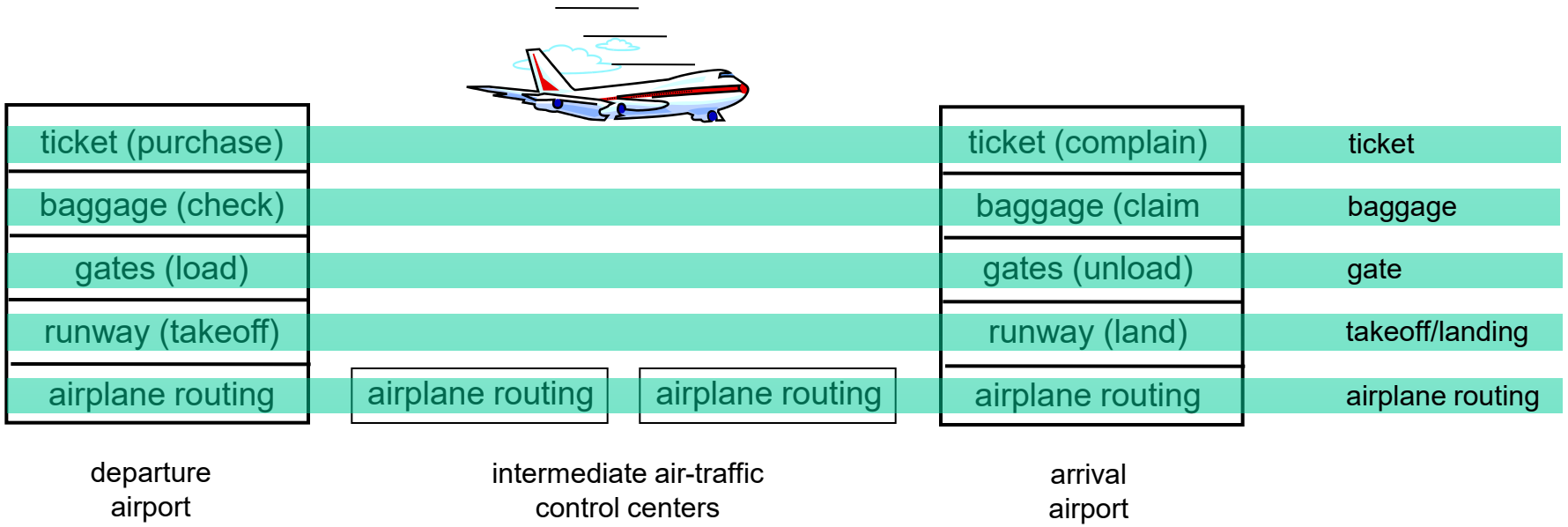
…. or at least our discussion of networks?

Organization of air travel



- a series of steps教材上乘机的例子

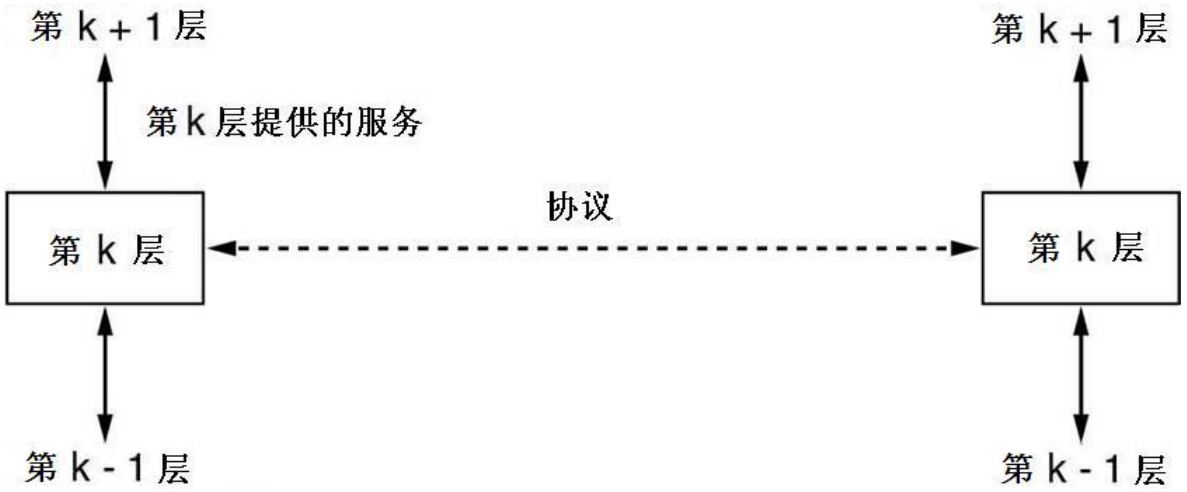
Layering of airline functionality



layers: each layer implements a service每一层都实现一项服务

- via its own internal-layer actions自己层的操作
- relying on services provided by layer below下面的层提供的服务

Layering of airline functionality



layers: each layer implements a service

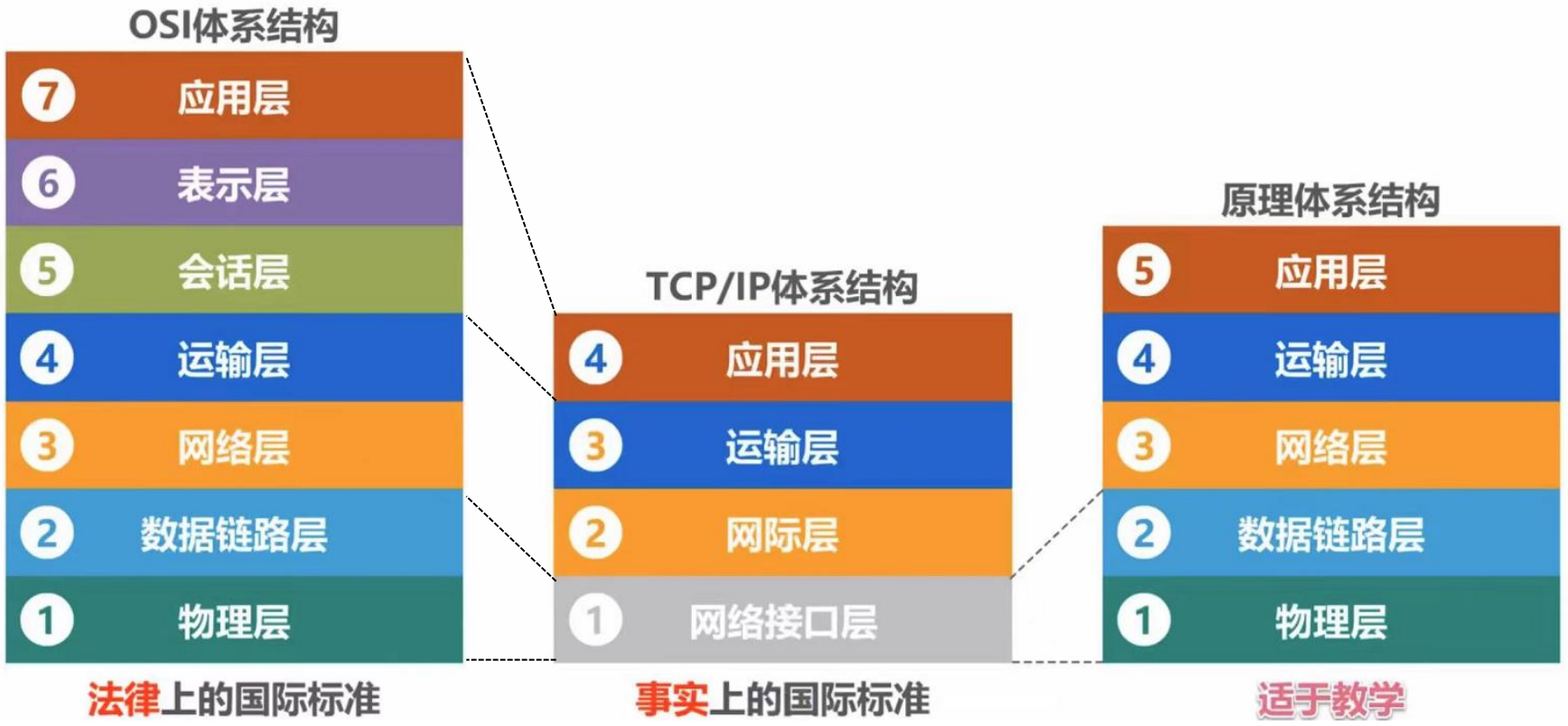
- via its own internal-layer actions
- relying on services provided by layer below

Why layering?

dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces通过明确的结构可以识别复杂系统的各个部分，并确定它们之间的关系
  - layered *reference model* for discussion供讨论的分层参考模型
- modularization eases maintenance, updating of system模块化便于维护更新系统
  - change of implementation of layer's service transparent to rest of system某层的服务执行改变不影响系统的其他部分
  - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful?

Network architecture

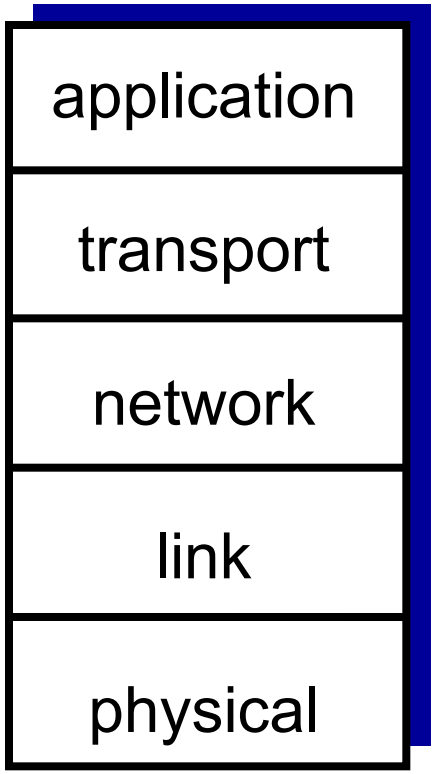


OSI/RM

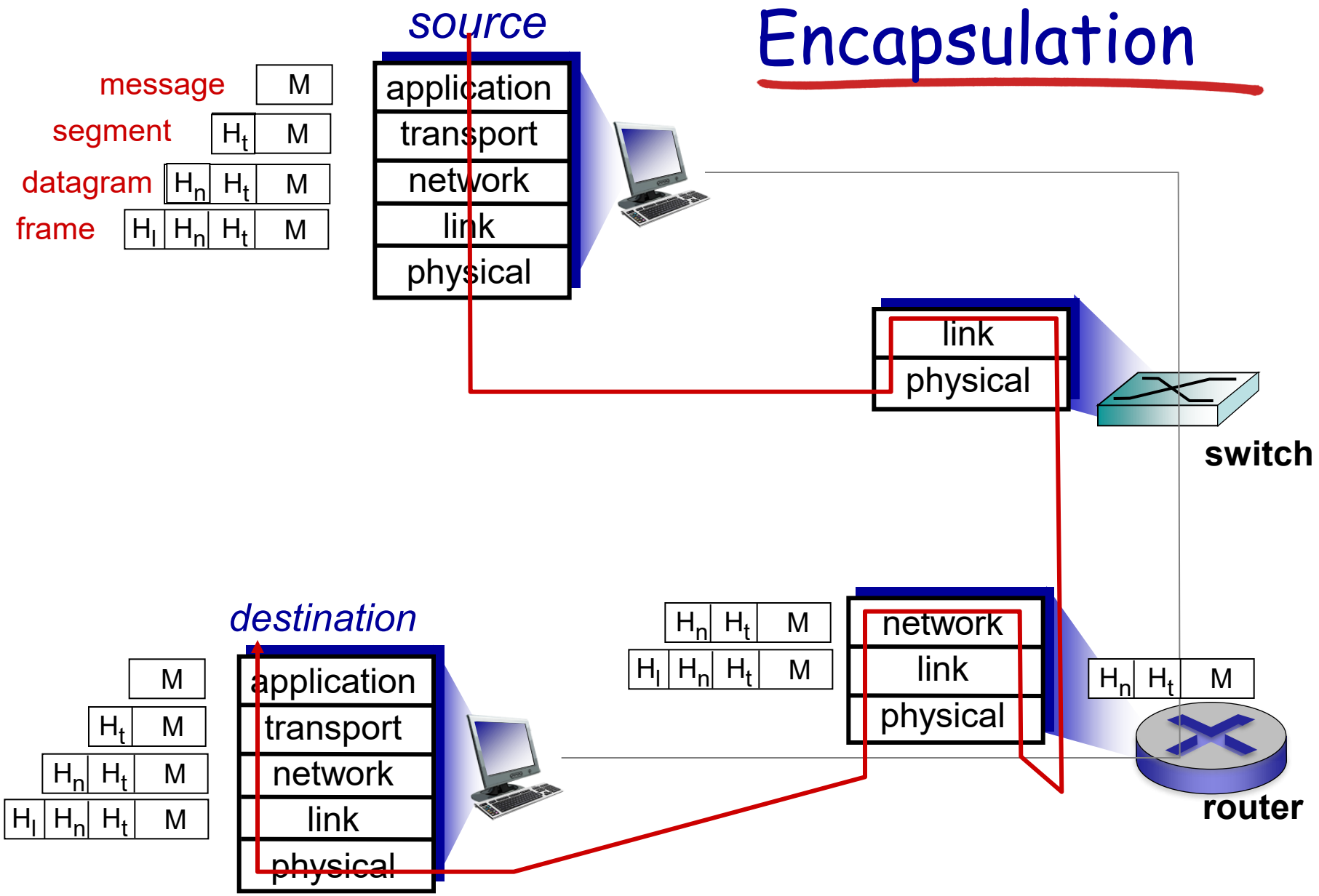
Internet RM

Internet protocol stack

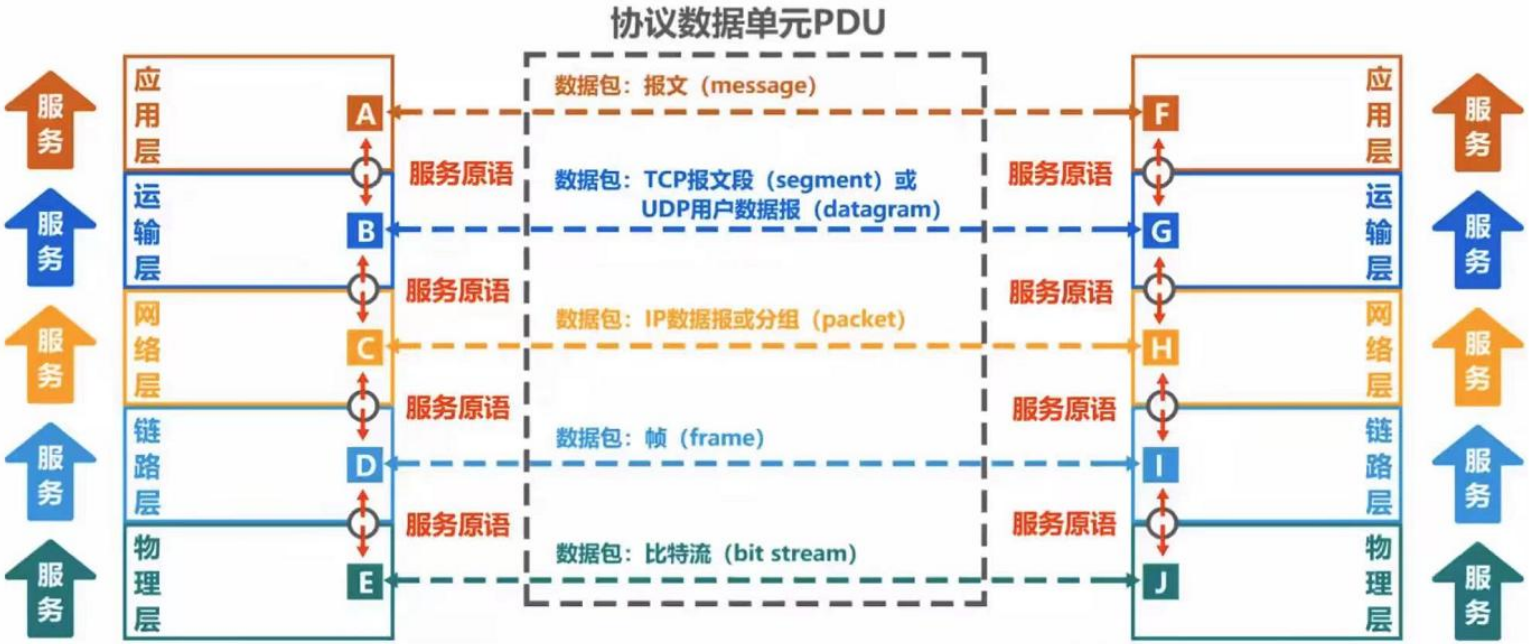
- *application:* supporting network applications
  - FTP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination将数据报从源站路由到目的地
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"



Encapsulation



Terminology of network architecture



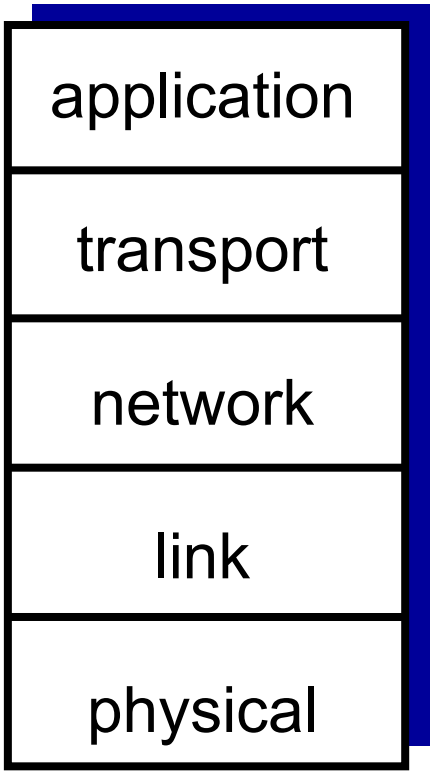
Entity: hardware and software process that can send and receive information. 实体：可发送和接收信息的硬件和软件过程。

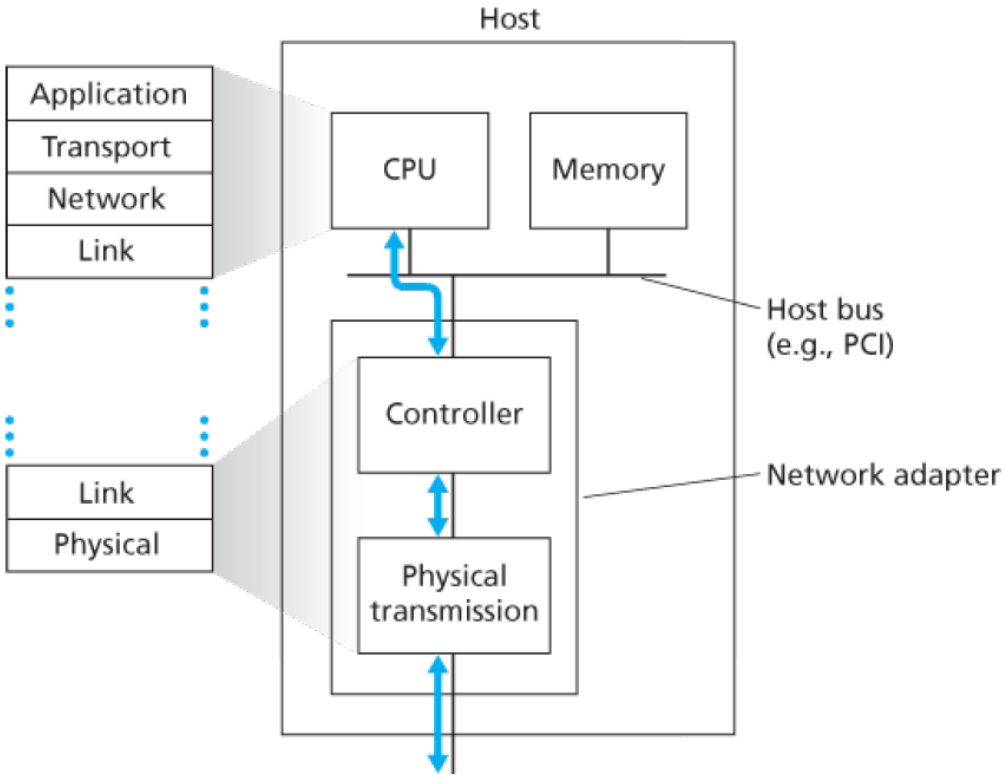
Service: the function provided by the lower layer to the upper layer低层提供给高层的功能

Service access point (SAP): places where entities of two adjacent layers interact服务接入点 (SAP)：两个相邻层的实体进行交互的地方

Service primitives: commands to be exchanged during the interaction between adjacent layers服务基元：相邻层之间交互时要交换的命令

Internet protocol stack





Exercise-4

1. In the OSI model, the first bottom-up level to provide end-to-end services is (B)

A. data link layer B. transport layer C. session layer D. application layer

2. Among the following options, which is not described in the network architecture is (C)

A. layers in the network B. protocols used by each layer

C. internal implementation details of the protocols

D. functions that must be completed by each layer

3. In the TCP / IP reference model, what directly provides services for the application layer is (D)

在TCP/IP参考模型中，直接为应用层提供服务的是

A. presentation layer B. transmission layer C. network layer D. network interface layer网络接口层

Exercise-4

4. Assuming that the application layer in the ISO/OSI reference model wants to send 400B data (without splitting), except the physical layer and application layer, all other layers introduce 20B additional overhead when packaging PDU, the transmission efficiency of the application layer is about ( )

A.80%　　B.83%　　C.87%　　D.91%

Assuming that the application layer in the ISO/OSI reference model wants to send 400B data (without splitting), except the physical layer and application layer, all other layers introduce 20B additional overhead when packaging PDU, the transmission efficiency of the application layer is about ( )
A.80%　B.83%　C.87%　D.91%

✓ 正在搜索: **ISO/OSI 参考模型 应用层 传输效率**

✓ 正在为你生成答案...

您好，这是必应。我很高兴为您解答这个问题。😊

根据**OSI参考模型**，应用层是最顶层，它负责提供网络服务给用户。OSI参考模型共有七层，除了物理层和应用层，其他五层都会在封装PDU时引入额外的开销[1][2]。物理层是最底层，它负责将比特流通过物理介质传输[3][4]。

假设除了物理层和应用层，其他五层每层都会引入20B的额外开销，那么应用层要发送400B的数据（不拆分），就需要封装成500B的PDU。因此，应用层的传输效率就是400B/500B，约等于0.8，也就是80%。所以，正确答案是**A**。

如果您想了解更多关于OSI参考模型或TCP/IP协议模型的信息，您可以访问以下链接[3][4]。希望这对您有帮助。谢谢！

Learn more　1 blog.csdn.net　2 nowcoder.com　3 zhuanlan.zhihu.com　2 共 30

Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

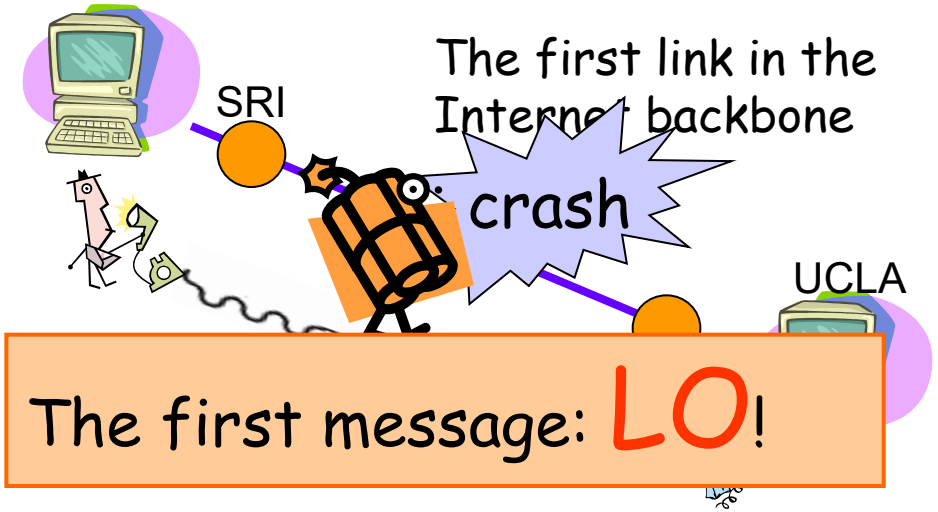
1.5 Protocol layers, Service models

1.6* Internet history

Internet History

1961-1972: Early packet-switching principles

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational



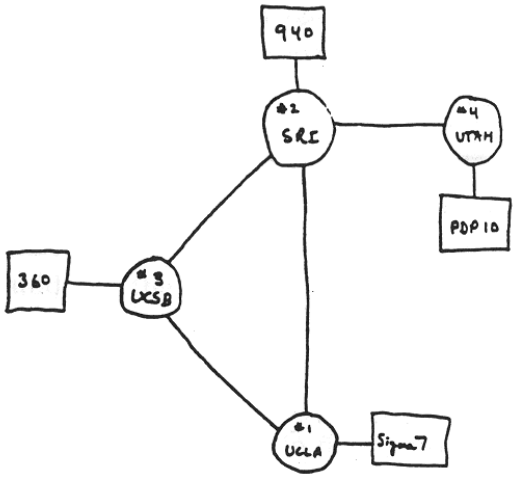
What was the first message ever sent on the Internet? (LOGIN)

- We sent an "L" - did you get the "L"? YEP!
- We sent an "O" - did you get the "O"? YEP!
- We sent a "G" - did you get the "G"?

Internet History

1961-1972: Early packet-switching principles

- The Internet is Born! at UCLA on October 29, 1969
- What it looked like at the end of 1969



THE ARPA NETWORK

- 1972:
  - ARPAnet public demonstration
  - NCP (Network Control Protocol) first host-host protocol
  - First e-mail program
  - ARPAnet has 15 nodes

Internet History

1972-1980: Internetworking, new and proprietary nets

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- ate70's: proprietary architectures: DECnet, SNA, XNA
- late 70's: switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

define today's Internet architecture

Internet History

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: SMTP e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: FTP protocol defined
- 1988: TCP congestion control

- New national networks: Csnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

Internet History

1990, 2000's: commercialization, the Web, new apps

- Early 1990's: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- Early 1990s: Web
  - Hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - Late 1990's: commercialization of the Web

Late 1990's – 2000's:

- More killer apps: instant messaging, P2P file sharing
- Network security to forefront
- Est. 50 million host, 100 million+ users
- Backbone links running at Gbps

Internet History

Recent Developments



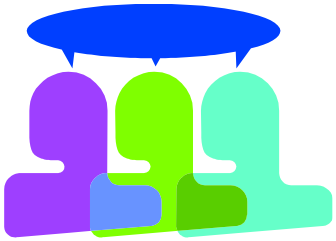


- High-speed access networking
  - 10Gbits Ethernet
  - Wireless access (UWB, WiMAX, 4G, etc.)



- Security
  - Internet
  - Wireless(5G)

- Other networking
  - Industrial internet
  - Internet of things
  - ……





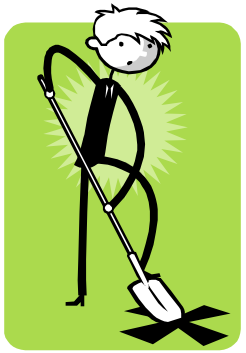
Introduction: Summary

Covered a "ton" of material!

- Internet overview
- What's a protocol?
- Network edge, core, access network
  - Packet-switching versus circuit-switching
- Performance: loss, delay
- Layering and service models
- History

You now have:

- Context, overview, "feel" of networking
- More depth, detail *to follow!*



Homework

- P95~97

R4,R13,R18
R23,R25

- P99~101

P6,P8,P10,P14